Stony Brook University

CSE 508

Network Security (Lecture 2- Background on Network and its Protocols)

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Modules

Stony Brook University

- This class will have the following modules:
  - Background:
    - Cryptography
    - Introduction to Network and Protocols
  - Authentication
  - Access Control
  - Attacks and Defenses against protocols
  - Essential tools for attack and defense
  - Web Security
  - Privacy and Anonymity
  - Advanced Topics (Protocol Verification)

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Acknowledgment

- Professor Douglas Jacobson's book inspired the content of the slides

- Book: "***Introduction to Network Security.***"

- The images used in this slide are directly from that book

Focus of the Lecture

- Introduction
- Layered Architecture
- Key Terminology
- Protocol Functions
- OSI Model
- TCP/IP Model
- Protocol Specification
- Protocol Addresses
- Protocol Headers
- The Internet
- Addressing
- Client Server communication
- Routing

History

Unfortunately, from the very beginning of the Internet, security has been a secondary concern

1840
- 1844 First Telegraph line
- 1861 Over 2200 telegraph offices
- 1866 First transatlantic cable
- 1875 First words on a telephone
- 1880 over 30,000 phones

1900
- 1900 over 600,000 phones
- 1910 over 5,000,000 phones
- 1920 over 11,000,000 phones

1950s
- Point-to-point network to main frames

1960
- 1968 300 baud modem
- 1969 ARPA NET (4 nodes)

1970
- 1971 15 nodes in APRANET
- 1973 TCP/IP development
- 1973 Ethernet was proposal in a Ph.D. Dissertation
- 1977 TCP/IP test bed
- 1979 UUCPnet

1980
- 1980 ARPANET virus (accidental)
- 1983 TCP/IP becomes the protocol for ARPANET
- 1984 over 1000 hosts
- 1986 NSFNET is started
- 1987 over 10,000 hosts
- 1988 Internet worm infects over 6,000 hosts
- 1989 over 100,000 hosts

1990
- 1991 WWW released by CERN
- 1992 over 1,000,000 hosts
- 1995 First ISPs started
- 1996 over 10,000,000 hosts

2000

Modularity in the Network Architecture

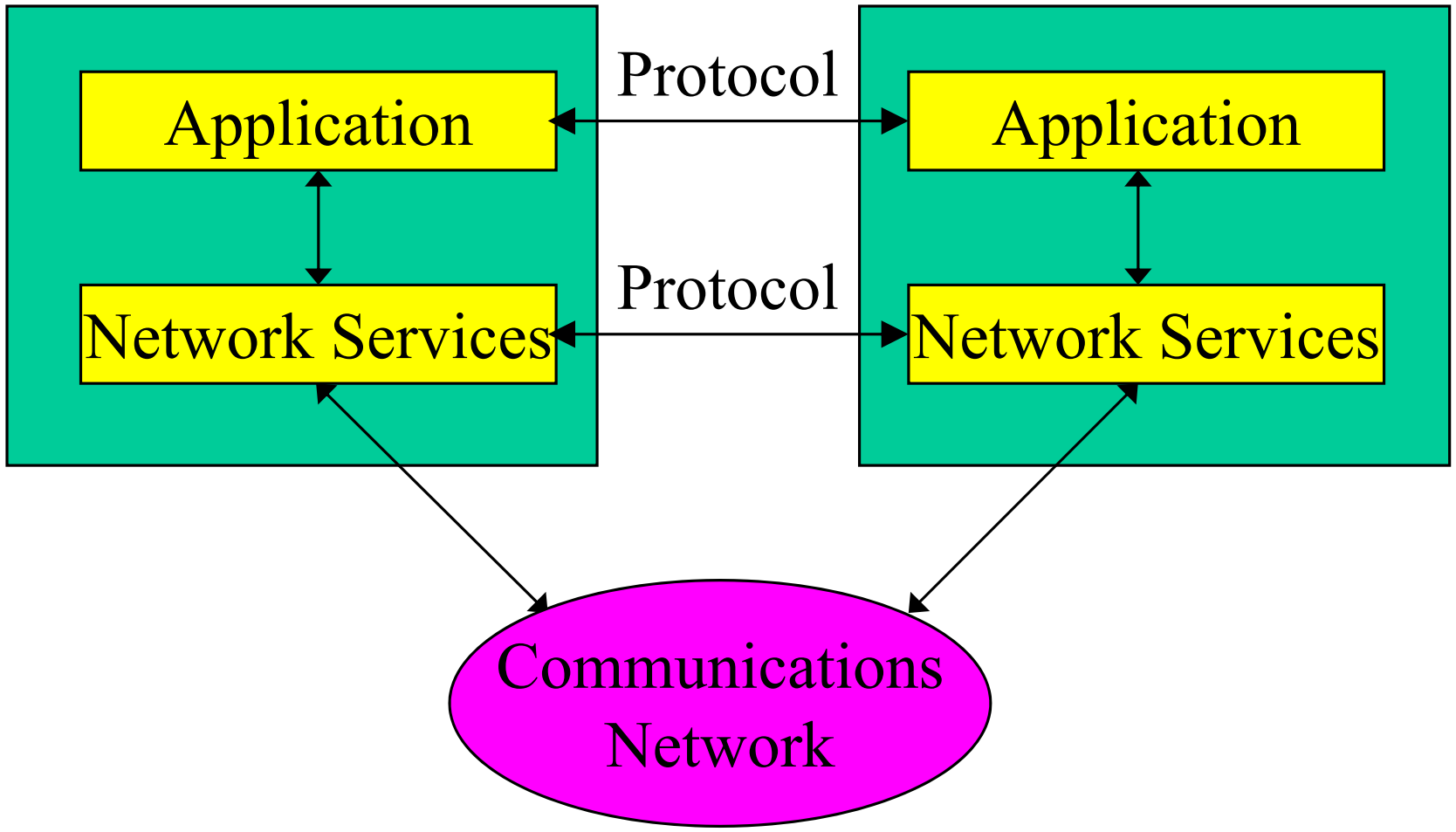
- The design of the network was done with modularity in mind.

- Why do we want modularity?
  - Separation of concerns
  - Simplification
  - Manageability
  - Reuse
  - Standardization

- This is akin to good practices in programming.

- As a network architect, you want developers writing apps to focus on the apps' behavior, not to worry about how to communicate over the Internet.

10,000 feet view of Layered Architecture



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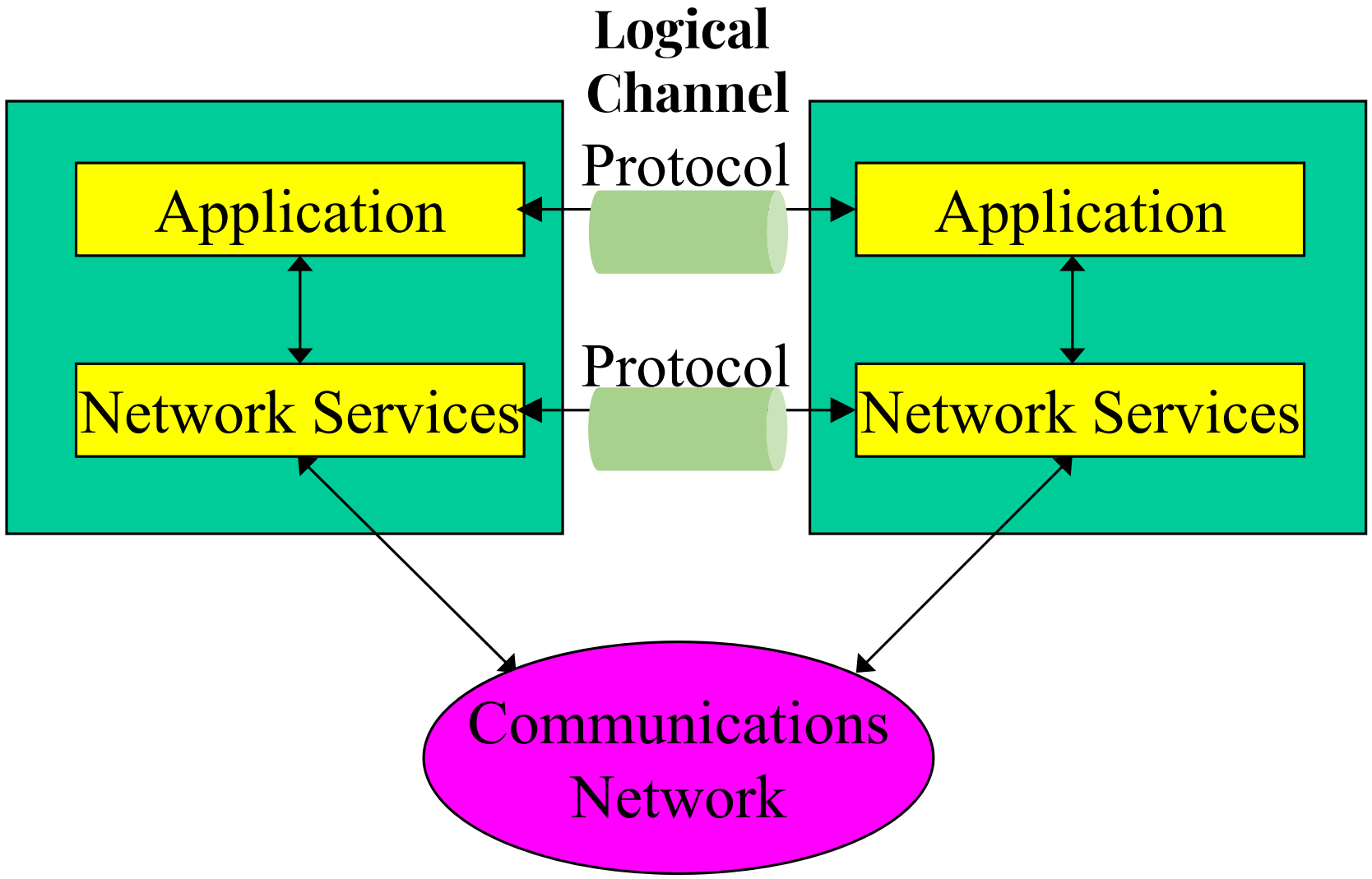

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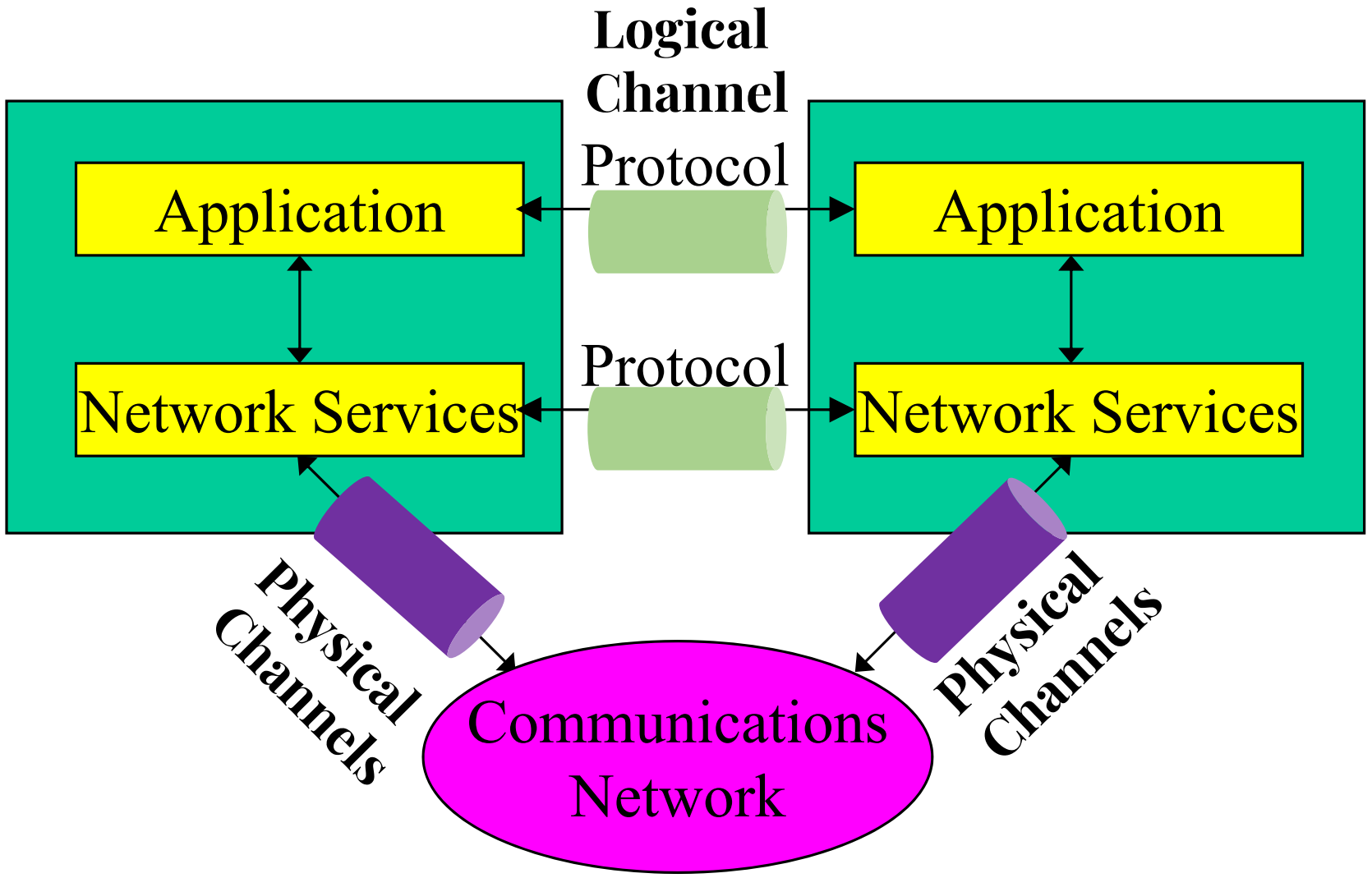

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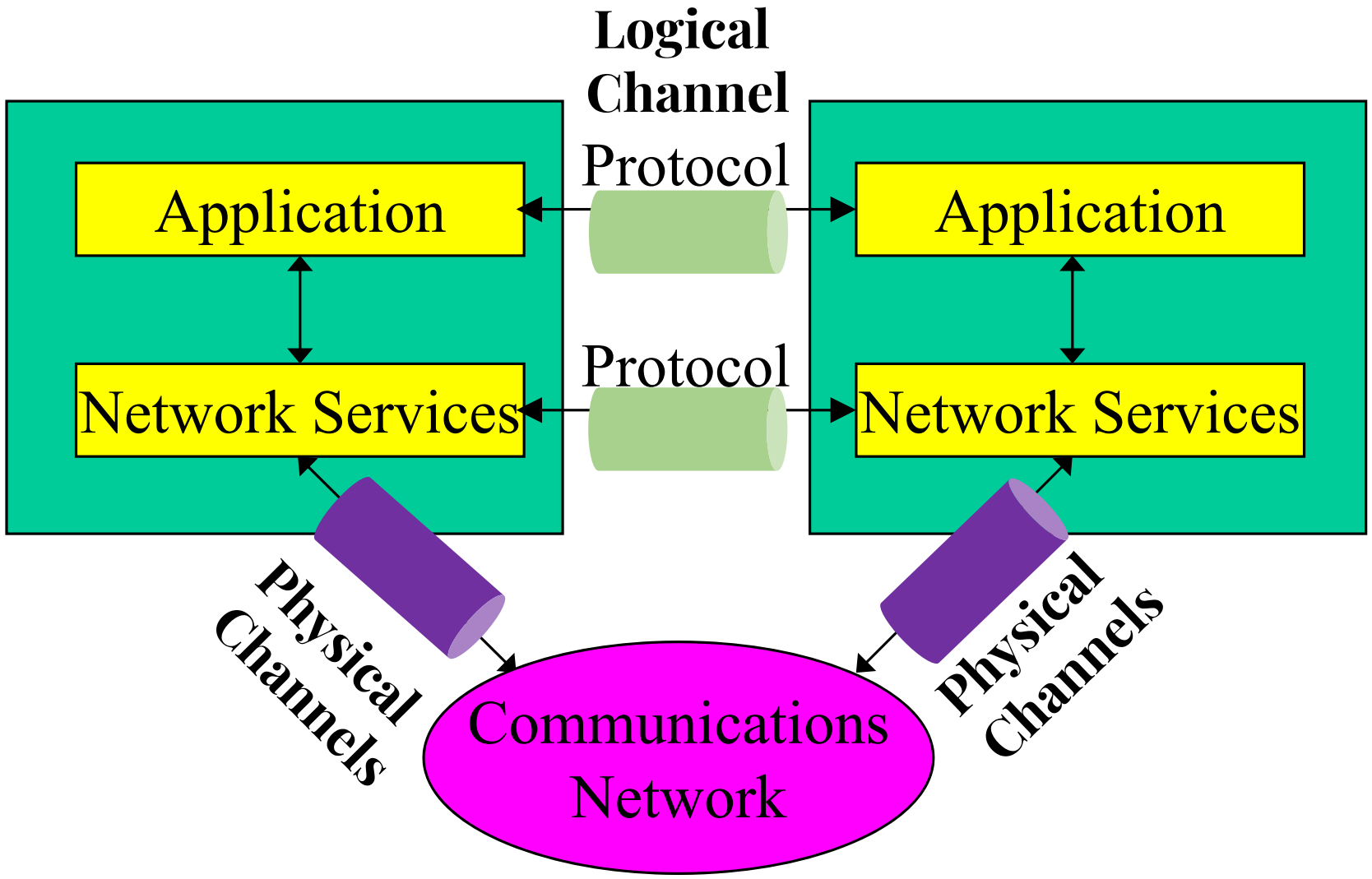
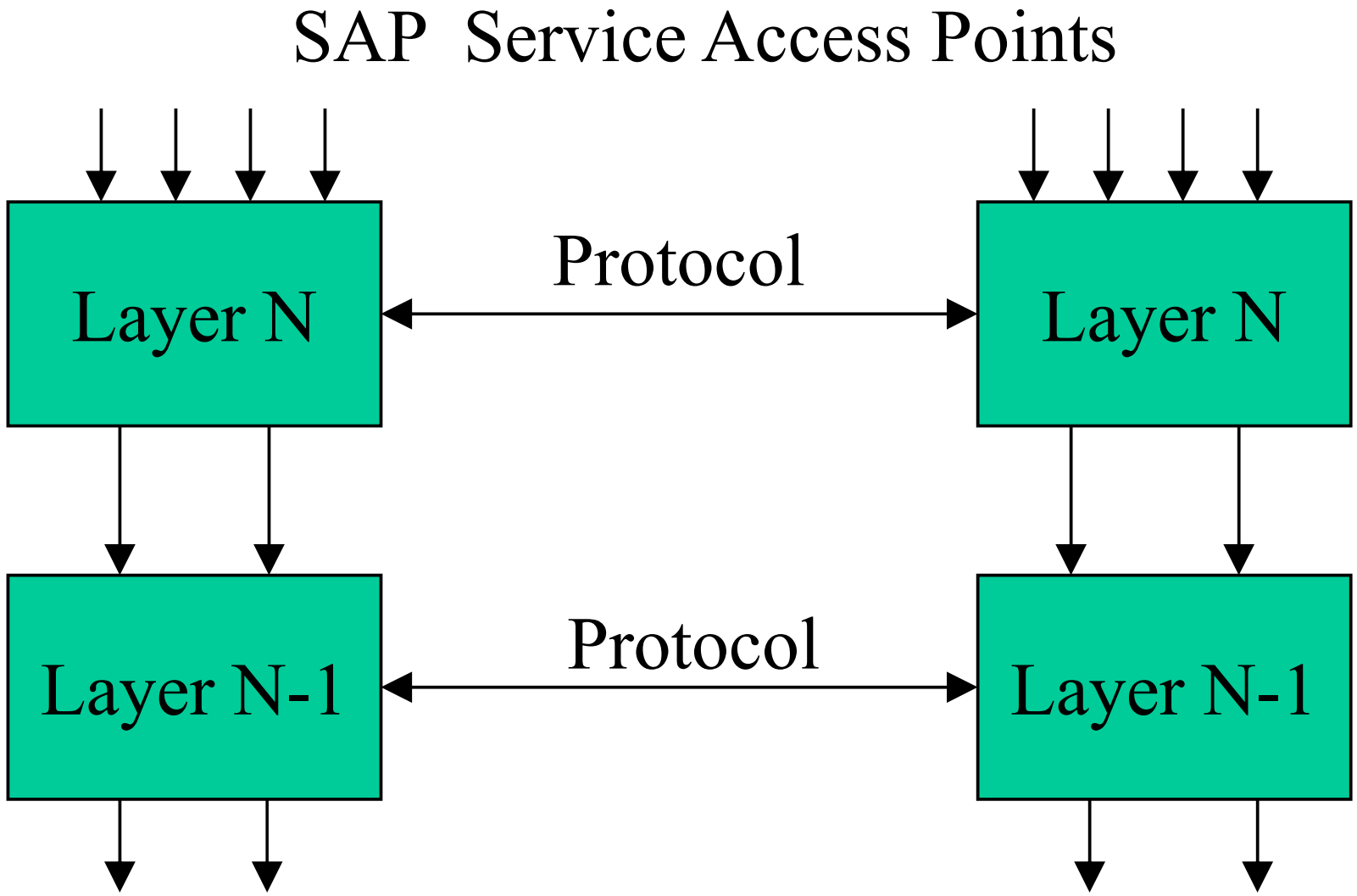


Application should work on both LAN and Wi-Fi

Principle of Layered Network Architecture



Need for Layered Architecture

- **Brought about because of a need for standards**

- **Layers:**
  - Take information from above (layer N-1)
  - and pass information below (layer N+1)

- The services are provided through the service access points (**SAPs**)

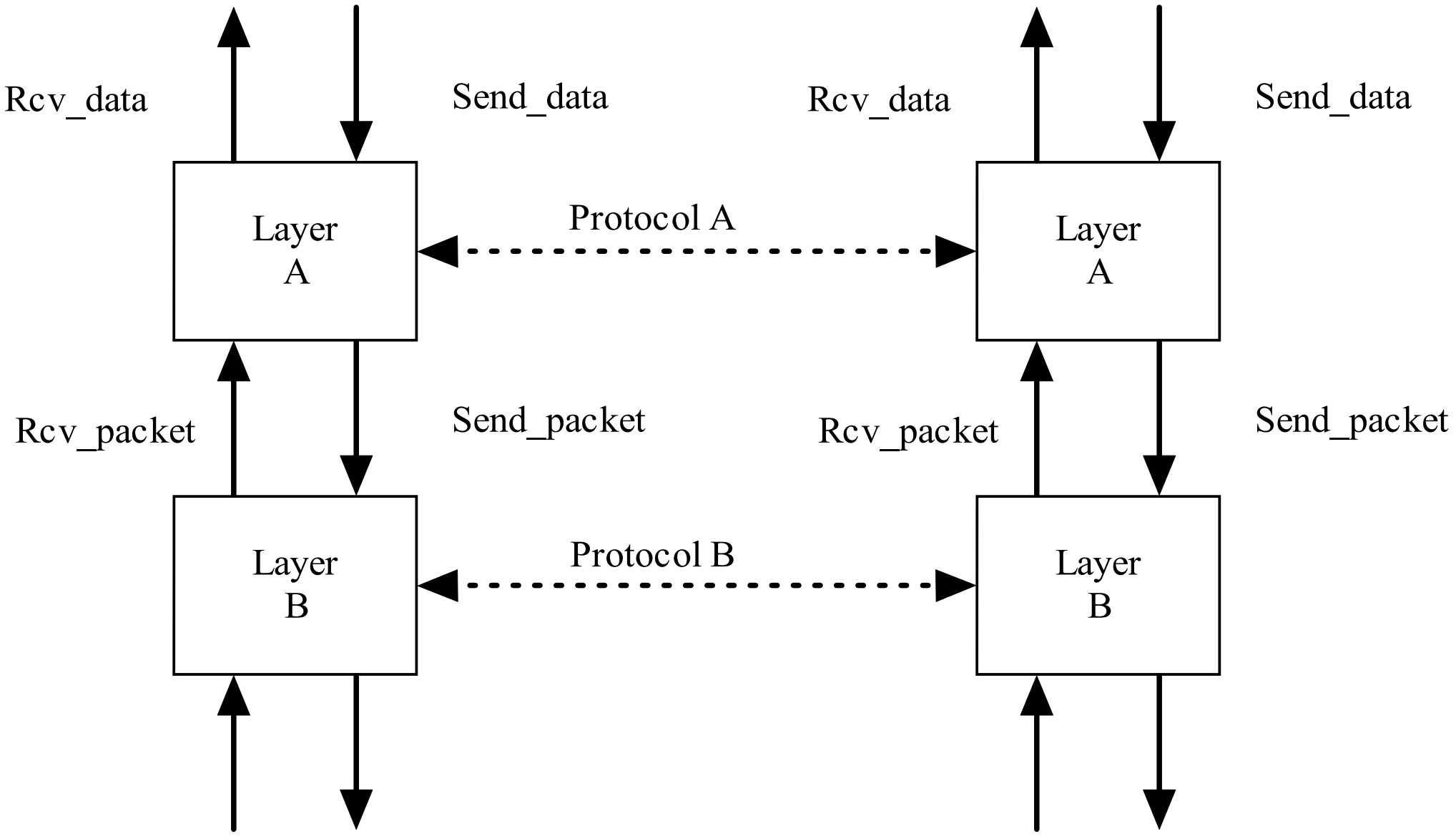
- Layer functionality is implemented through an entity

- Each layer contains one or more entities that are responsible for providing services to the N+1 layer

Layered Architecture (logical & physical channels)

- In order for layers to carry out functions, they need to communicate
- A layer **N** entity may need to communicate with another layer **N** entity, which does not reside on the same system, to provide the service.
- The layer **N** entity uses the layer **N-1** services to communicate with the remote layer **N** entity.

Layered Architecture (logical & physical channels)



What are Protocols?

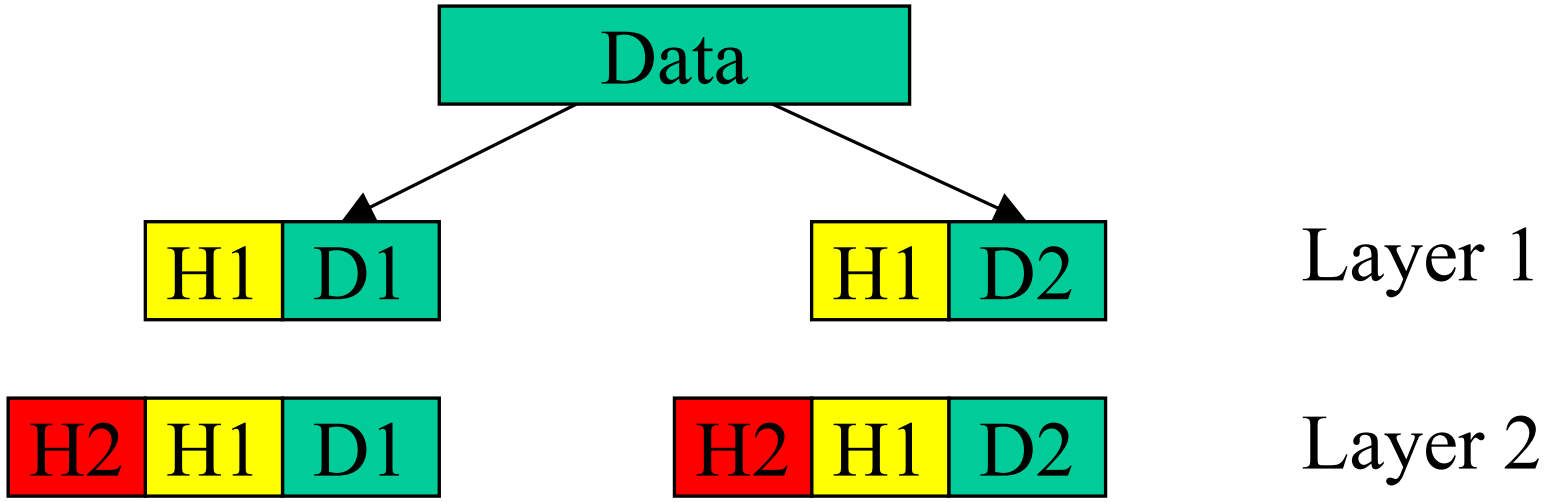
- **PROTOCOLS are the <u>rules/discipline</u> that have been defined for the layer N to layer N communication.**
- They represent extra information
  - Example: saying "hello" on the telephone is a protocol
- Protocols indicate when to send data, what language to use, etc.
- A layer specification defines
  - ***What protocol does it use?***
  - ***What does it expect as input (SAPs)?***
  - ***What functions does it provide?***
- Layer specifications allow multiple vendors to have the same functionality.
  - **Example:** Different ethernet card brands

Protocol Data Unit (PDU)

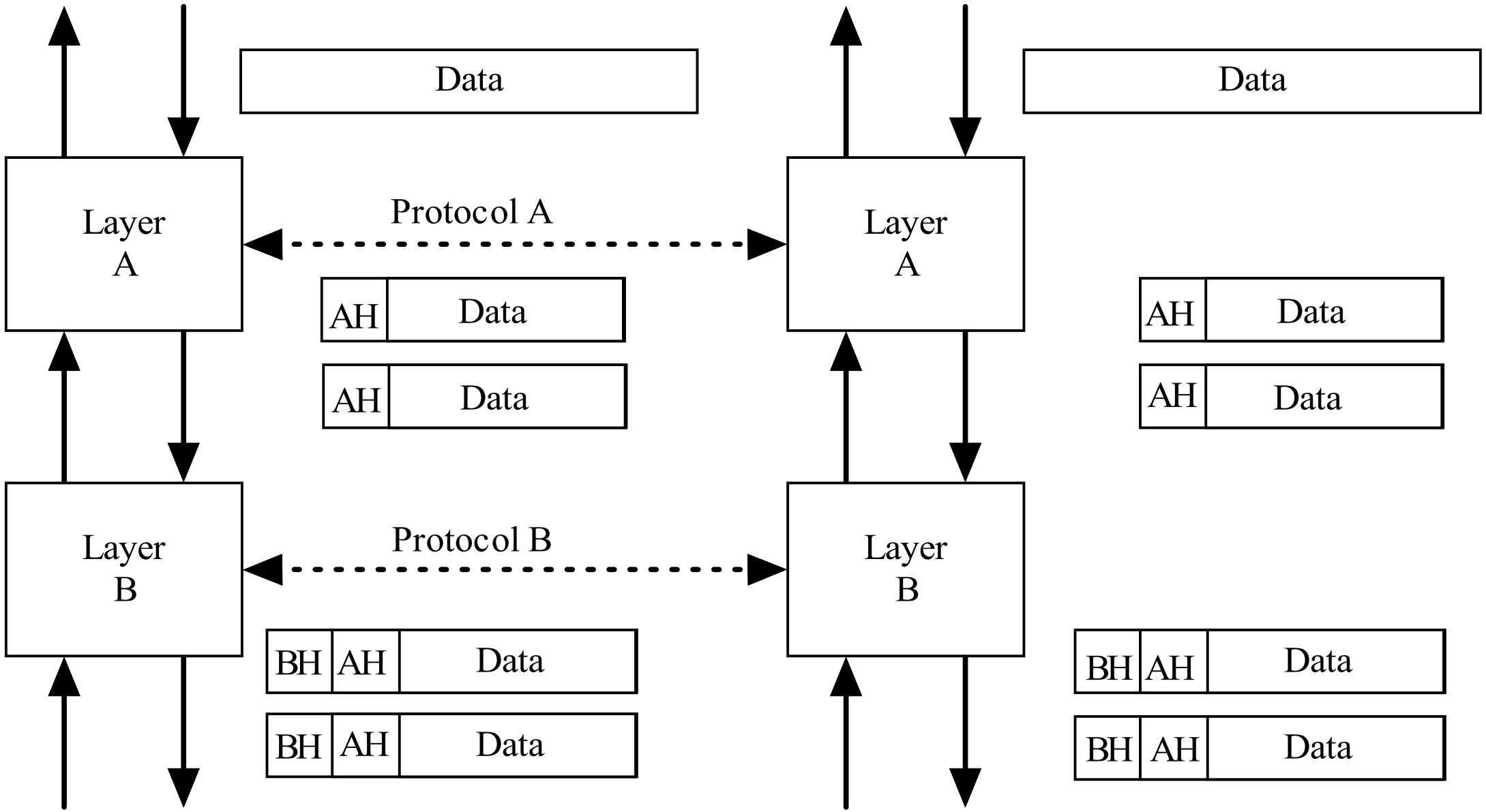
- **Protocol Data Unit (PDU)** is the combination of data from the higher layer and the protocol or control information.

- The protocol or control information created by a layer is called the **header**.

- **Each layer adds its own header**



Layered Architecture (with PDU)



Key Terminology we have seen so far

- The ***protocol*** defines the rules/discipline for PEER entity communication
- ***Service Access Points (SAP)*** specify how the N entity communicates with the N-1 entity.
- ***Services*** are provided by the N entity to the N+1 entity
- ***Functions*** are provided by the entity in coordination with the peer entity.

Basic Functions of a Protocol in Networking

1. Segmentation and reassembly
2. Encapsulation
3. Connection control
4. Ordered delivery
5. Flow control
6. Error control
7. Multiplexing

Segmentation and reassembly

- Often, physical media or error control issues dictate a maximum data size.
  - Generically, this size is called ***Maximum Transmission Unit*** (**MTU**)

- Therefore, the data must be divided into smaller packets (***Segmentation***)

- And eventually put back together (***Reassembly***)

- ***Reassembly instructions are included in the header***

Encapsulation

- Encapsulation is the addition of control information to the data element in the form of a header.
  - **Address:** The address of the sender and/or receiver.
  - **Error Detection Code:** Some sort of code is often included for error detection.
  - **Protocol Control:** Additional information that is needed to implement the protocol.

Connection Control

- **Connectionless Data Transfer**
  - Data is transferred without prior coordination
  - No set path
- **Connection-oriented Data Transfer**
  - A logical association, or Connection, is established between entities before any data is transferred
  - **Example:** telephone
- Three phases of Connection Control
  - ***Request/connect phase***
  - ***Data transfer phase***
  - ***Termination phase***

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Connection can be viewed as a communication state

Ordered Delivery

- Pieces arrive in the same order as sent
- **Not provided by connectionless protocols**
- Not required to be provided by Connection-oriented protocols, but it is common for most. (needed for file transfer)

Flow Control

- *Technique for assuring that the transmitting entity does not overwhelm a receiving entity.*
- Flow Control is typically implemented in several layers.
- Sometimes, the goal of flow control is to ensure **fairness**.

- **Flow control is found in most connection-oriented protocols.**

Error Control

- *A technique that allows a protocol to* ***recover*** *from* ***lost or damaged*** *PDUs.*

- *Three mechanisms:*
  - ***Positive acknowledgment***
  - ***Retransmit after timeout***
  - ***Error detection***

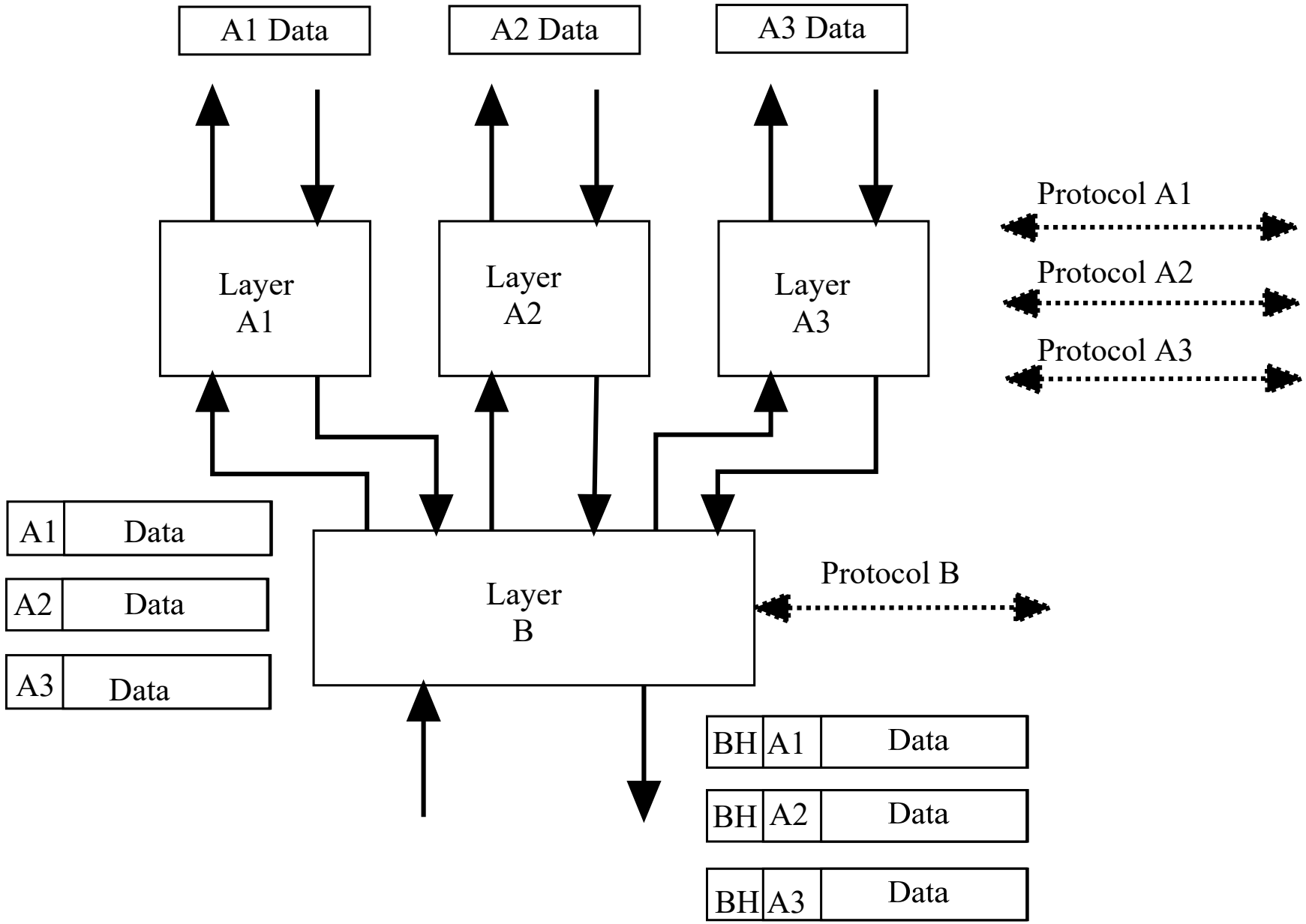
Multiplexing

- ***Upward Multiplexing*** *occurs when multiple higher-level connections are multiplexed on a single lower-level connection.*
  - ***Example:*** *Many applications (telnet, FTP, email) utilize TCP*

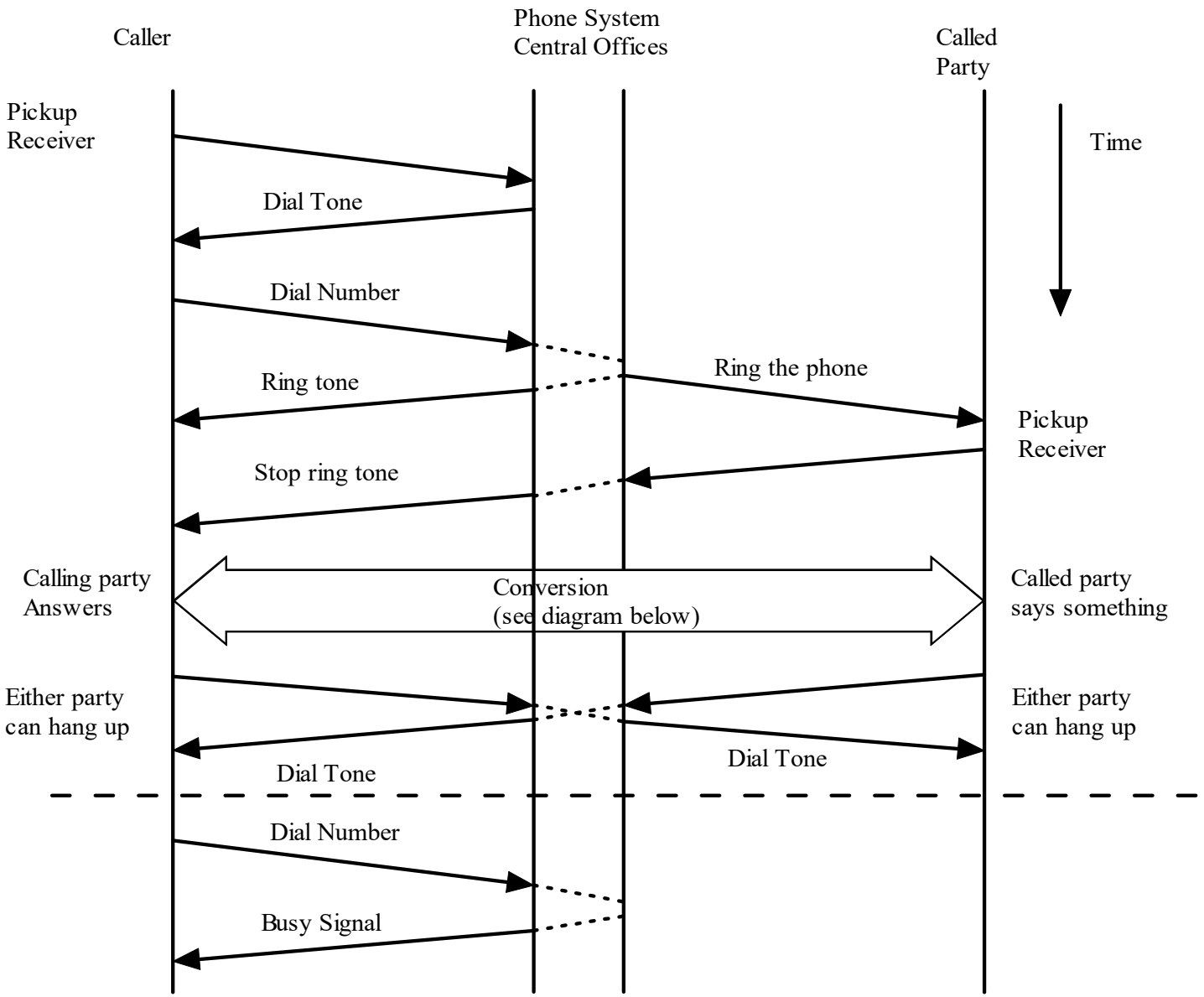
- ***Downward Multiplexing*** *occurs when a single higher-level connection is multiplexed on multiple lower-level connections. (***not as common***)*

- ***Addressing is needed to support multiplexing****.*

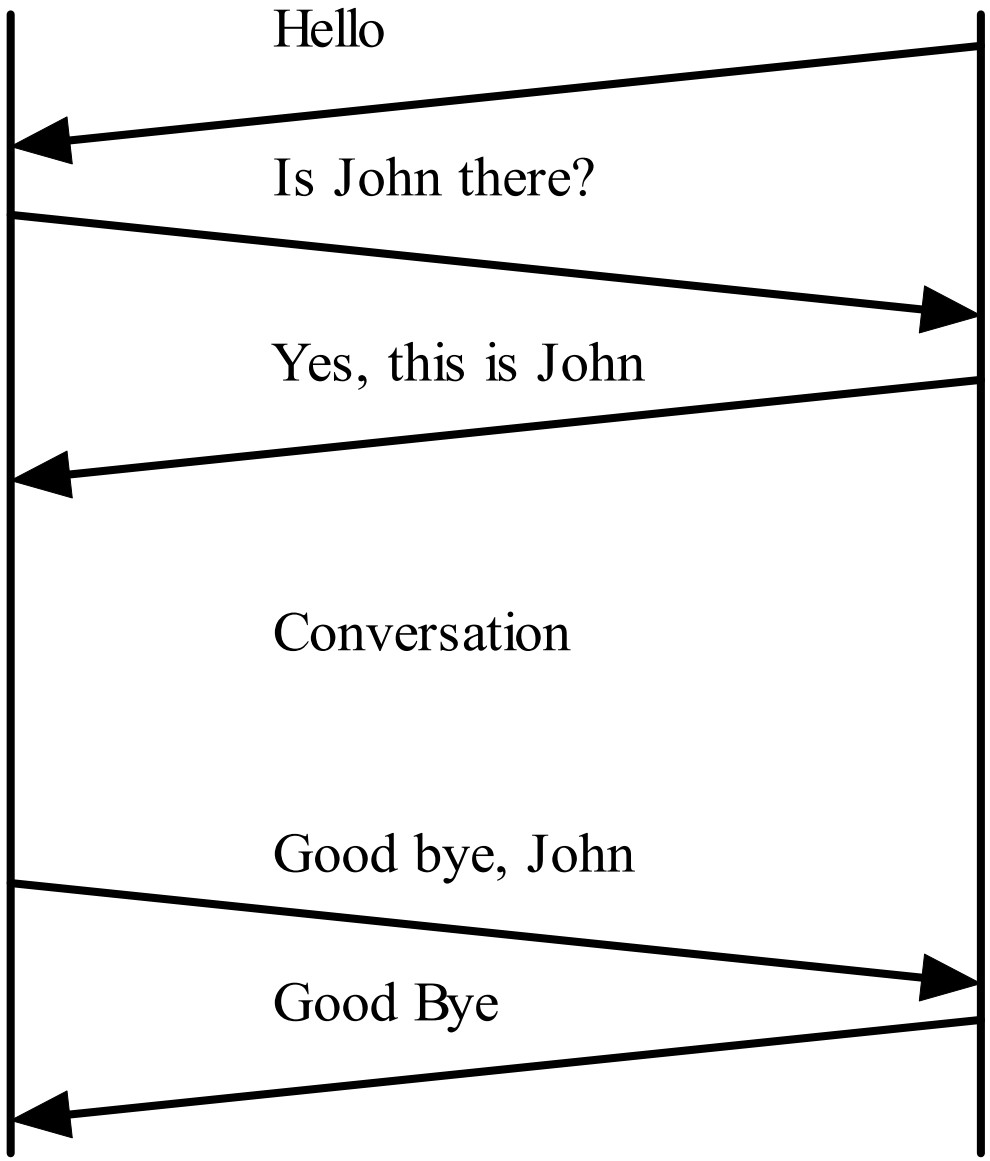
Multiplexing



Protocol Example (Lower Layer)



Protocol Example (Higher Layer)



OSI Model of Network

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical

Physical Layer

- Responsible for the transparent transmission of bit streams across the physical interconnection of systems
- Two configurations:
  - **Point-to-point**
  - **Multipoint**
- Physical layer must provide the data link entities with a means to identify the end point.
- Physical connection can be **Full Duplex** or **Half Duplex**
- Physical connection can be either **bit serial** or **N bit parallel**
- Physical layer must **deliver the bits in the same order** in which they were offered for transmission by the Data Link Layer.

Data Link Layer

- **The main task is to shield the higher layers from the characteristics of the physical transmission medium.**
- It should provide the higher layers with a reliable transmission that is basically **Error-Free**, although errors may occur in the transmission on the physical connection.
- Services provided should be independent of the data transmitted.
- The data link layer connects two network entities in adjacent systems called **Data link connections**.

Data Link Layer (contd.)

- Each data unit from the network layer is mapped into the data link protocol data unit along with the data link protocol information and is called a **Frame**.

- The data link layer must provide a method of ***recognizing the start and end of the Frame***.

- **Frames must be presented to the network in the same order they are received.**

- The data link layer should also implement **Flow Control** to prevent data overrun.

Network Layer

- The primary responsibility of the network layer is to provide the **transparent transfer of all data submitted by the transport layer** to any transport entity anywhere in the network.

- **The network layer must handle the routing of data packets.**

- **The network layer can be the highest layer in a device, such as a gateway or router.**

- IP protocol

Transport Layer

- **Responsible for a reliable transparent data transfer between two session layer entities.**

- Transport connection is provided to the session entities independent of their location.

- **The transport layer must optimize resources while maintaining a guaranteed quality of service.**

- The session layer requests a level of service, and once the transport connection is provided with a certain quality of service, it must be maintained unless it is notified of the change.
- **TCP protocol** (connection-oriented), UDP (connection-less)

Transport Layer (Contd.)

- The transport layer is only concerned with the transfer of data between session layers. It is not aware of the structure of the underlying layers or the topology.

- The transport layer uses the network layer to establish a network connection from one transport entity to another.

- Depending on the quality of the network, the transport layer may have to perform additional functions to offer the service.

- **The transport layer provides flow and error control.**

Session Layer

- The session layer is not concerned with the network.

- **The session layer's goal is to coordinate the dialog between presentation layers**

- **The session layer must provide the establishment of a session connection and the management of the dialog on that connection.**

- **Example:** An atm maintains a constant connection with a bank (transport service). **The session starts when the user logs in.**

Presentation Layer

- The presentation layer provides the application layer with services related to the presentation of information in a form that is meaningful to the application entities.

- The presentation layer provides the mechanism for the application layer to translate its data into a common format that can be translated by the peer application entity.
  - ***Marshalling*** and ***Unmarshalling***
  - ***Data format M1 → Network format → Data format M2***

Application Layer

- The highest layer provides a means for application processes to access the OSI stack.
- Provides both general services and application-specific services.
- This is what the user sees and the developers interact with in most cases
- **Examples:** telnet, ftp, web

Detailed View of OSI Model of Network

User

- Application
- Presentation
- Session

OS

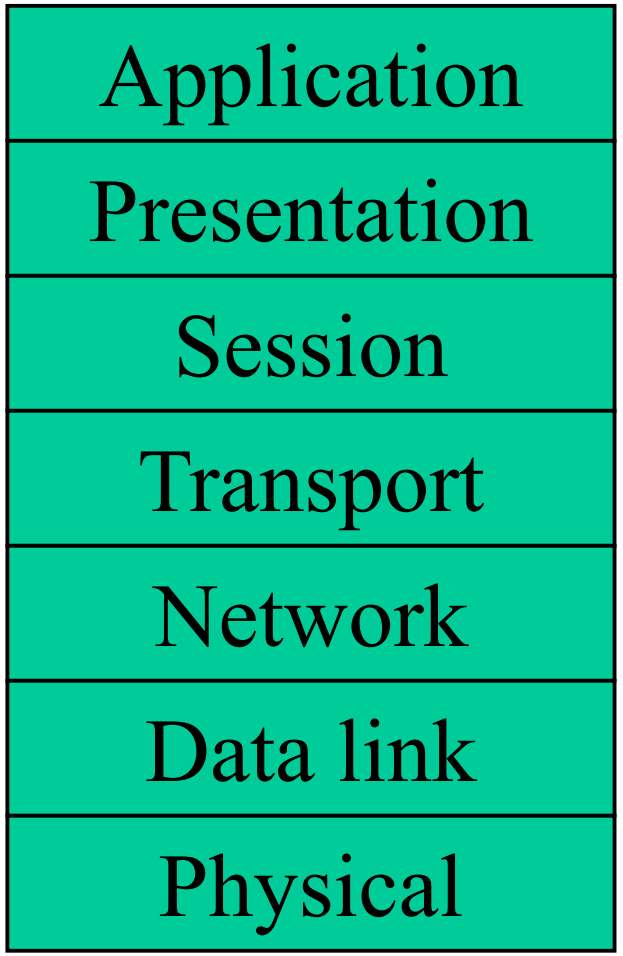
- Transport
- Network
- Data Link
- Physical

software

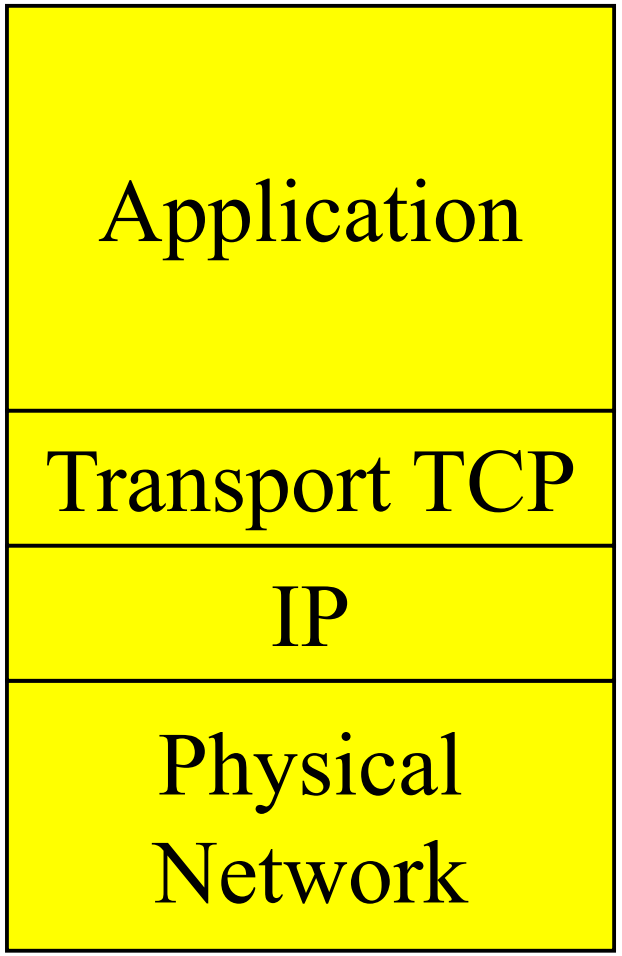
firmware

hardware

OSI 7-Layer Model and TCP/IP 4-Layer Model



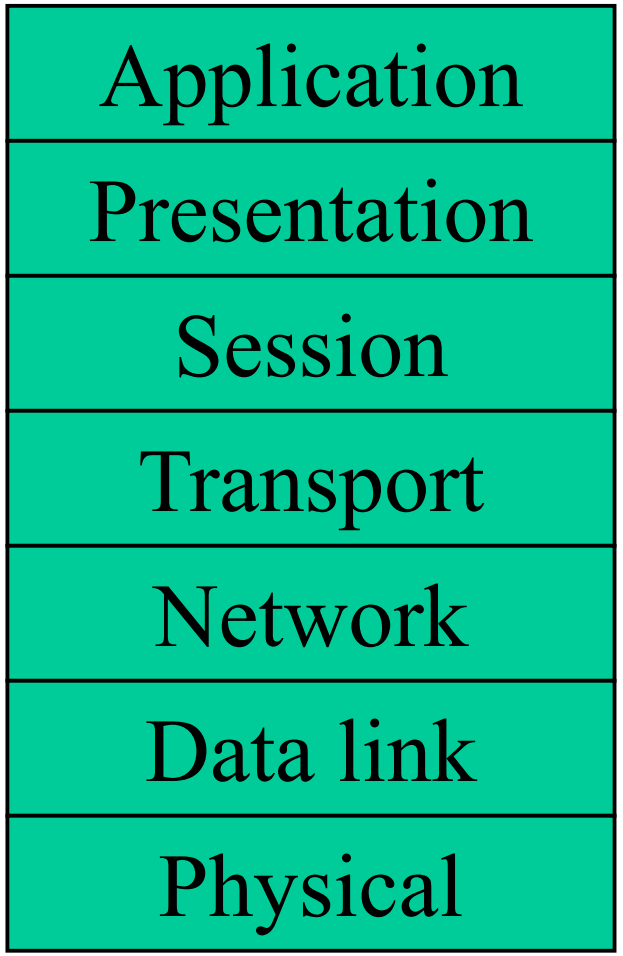
OSI 7-layer Model



TCP/IP 4-layer Model

OSI 7-Layer Model and TCP/IP 4-Layer Model

A standard heavily inspired by telecommunication industry



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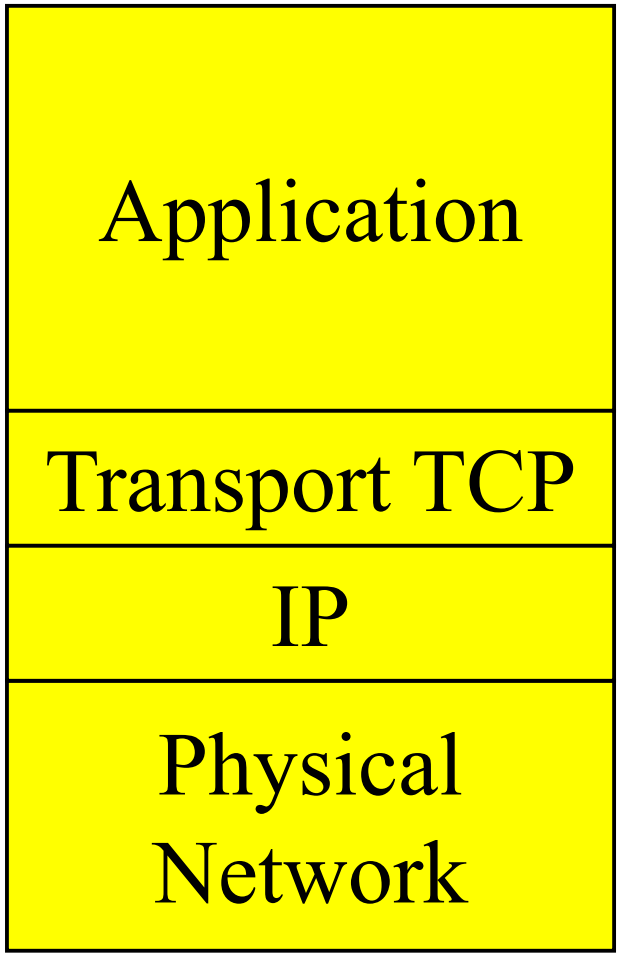

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Application
Presentation
Session
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OSI 7-layer Model

Application
Transport TCP
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Physical Network

TCP/IP 4-layer Model

V. Cerf and others proposed packet switching, which inspired TCP/IP in 1970s

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Abandoned

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TCP/IP 4-layer Model

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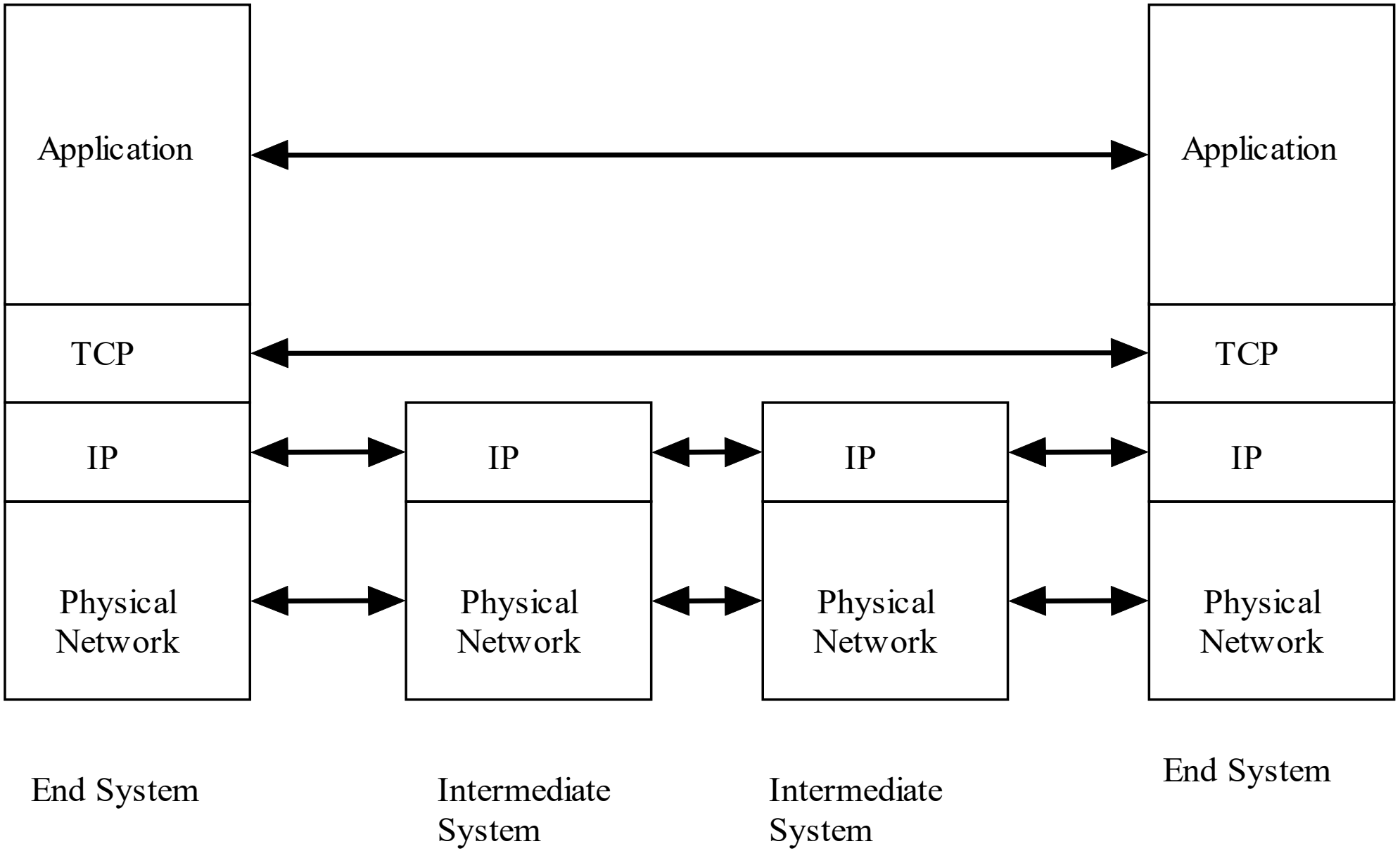
Our Internet uses TCP/IP

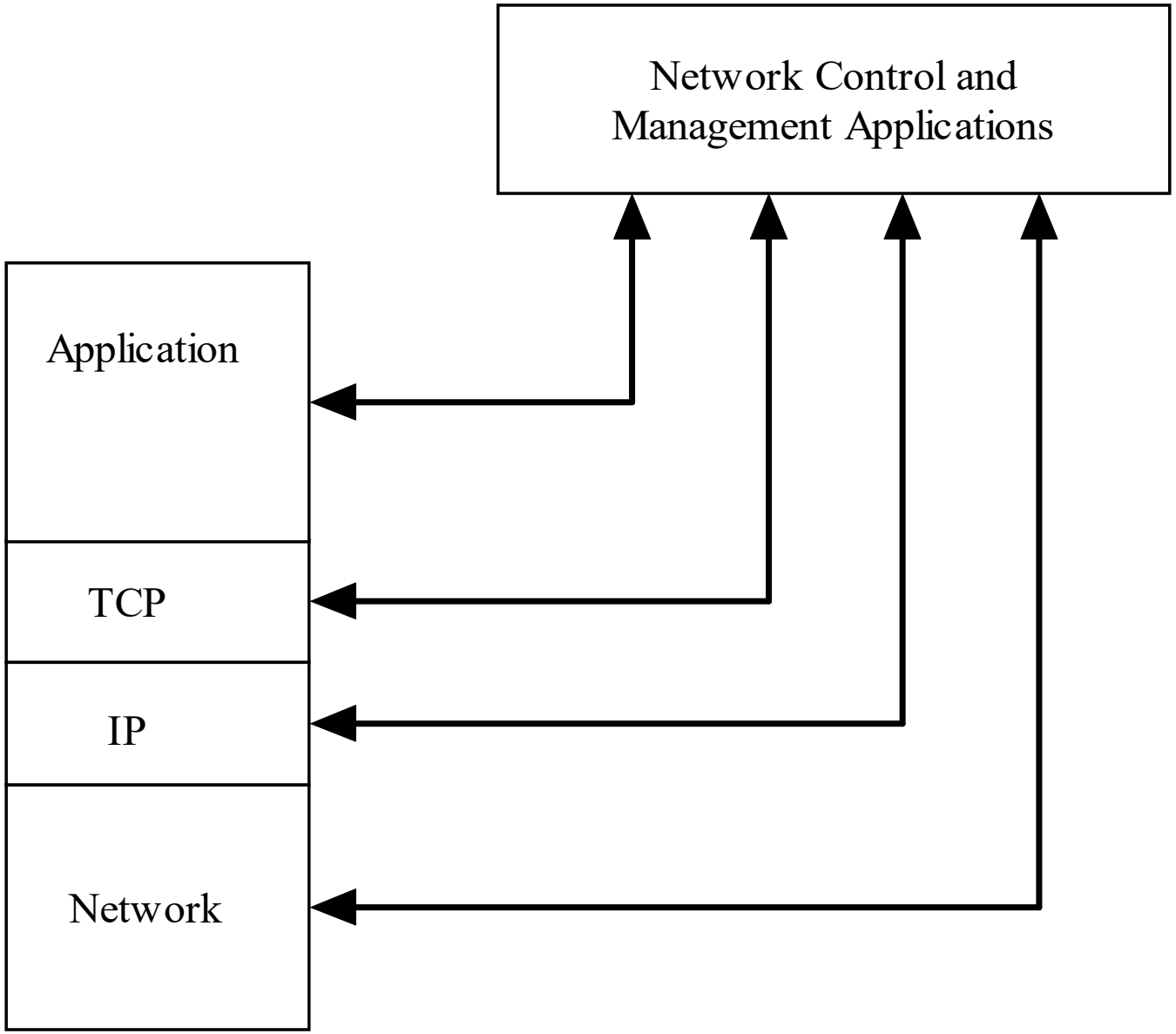
TCP/IP 4-Layer Model



Network Management, Monitoring, and Security

- When the layered architecture was designed, little to no attention was given to network management, monitoring, and security.
- These services were not considered critical when networks were small and controlled only by a few organizations.
- With the growing size and complexity of networks and the Internet, the need for these services became critical
- **Looking through the services, it became clear that the services require access to all the layers of the network to make reasonable decisions.**
- Due to these observations, a new architecture had to be considered where layered and non-layered services are combined.

Network Management, Monitoring, and Security



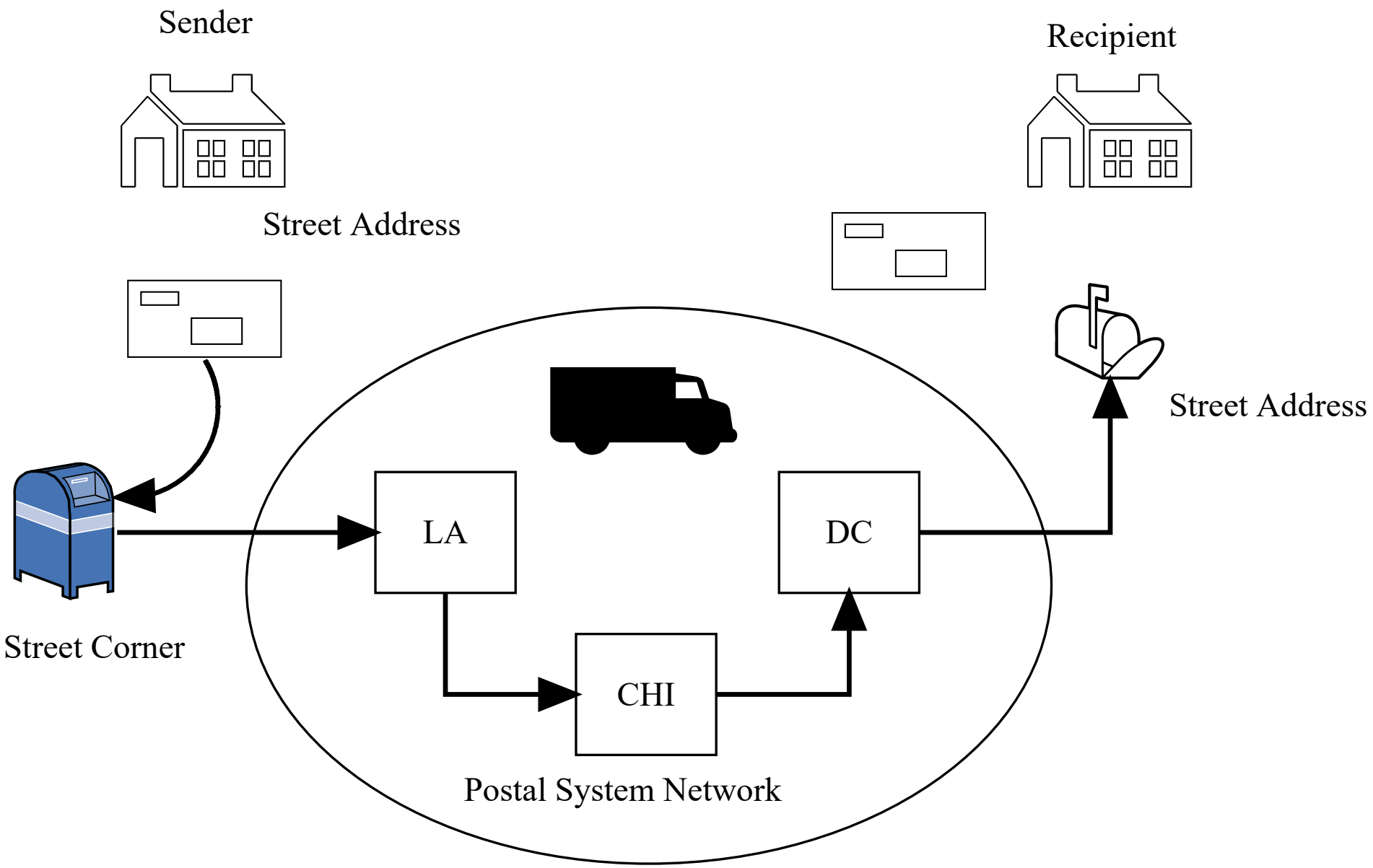
Protocol Specifications

- Open vs. Closed
- Specification methods:
  - English descriptions (RFC)
  - Flow and timing diagrams
  - Mostly incomplete, because only shows good flows and does not consider error cases
- Implementation flaws

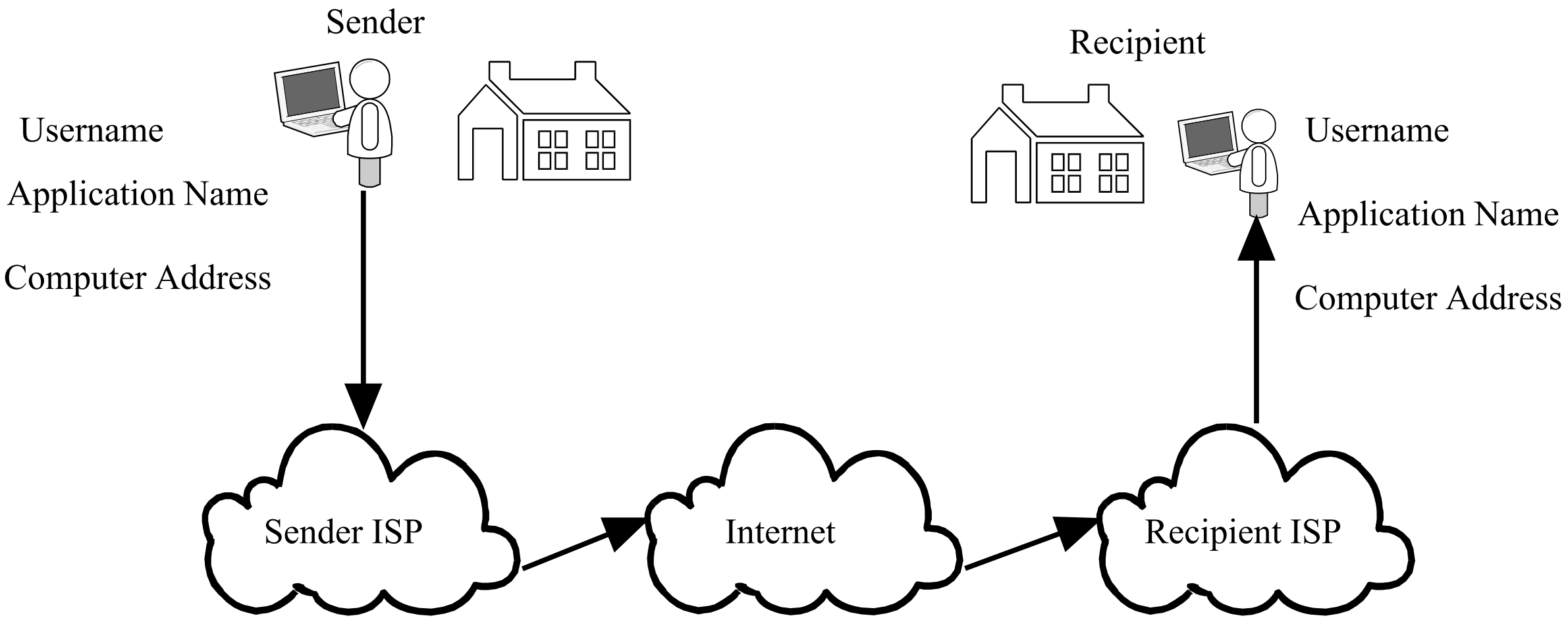
Network Standards

- Services provided
- Services expected
- Functions provided
- Protocol and packet formats
- Timing and sequence of packets

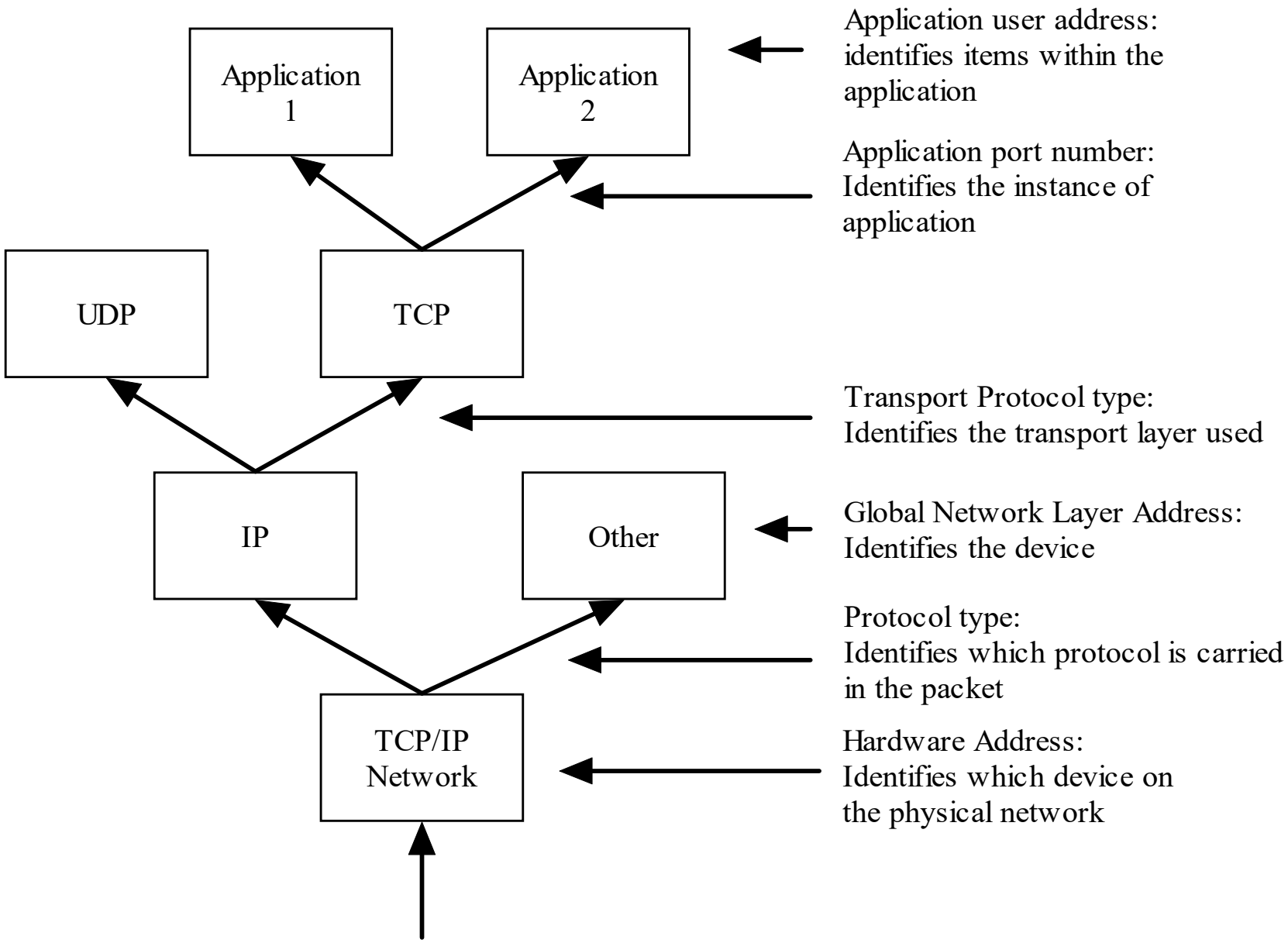
Need for Addressing



Need for Addressing in the Internet



Addresses


Application 1
Application 2
UDP
TCP
IP
Other
TCP/IP Network
Application user address: identifies items within the application
Application port number: Identifies the instance of application
Transport Protocol type: Identifies the transport layer used
Global Network Layer Address: Identifies the device
Protocol type: Identifies which protocol is carried in the packet
Hardware Address: Identifies which device on the physical network

Address Assignment

- Static
  - Configuration
  - Built-in
- Dynamic
  - Protocol discovery
  - User provided
- **Addresses can be assigned by:** Central Authority; Ad-hoc; Locally

Address Assignment

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Hardware addresses are assigned by vendors used for filtering but can be changed

IP Address Assignment

- Global address allocation
- Address assignment
  - Protocol-based (DHCP)
  - Static
  - Locally controlled
- Addresses can be changed

Application Address Assignment

- Port Number (Much less control)
  - Well known ports
  - Protocol-based discovery
  - Configuration based
  - User-input based

Hostname Address Assignment

- Often political and/or commercially-driven
- Assignment via central authority
- Protocol to find the IP address of a given name
  - We remember www.google.com, and not necessarily the IP address of Google

Protocol Headers

- **Fixed packet type**
  - Easy to parse
  - Limited functionality
- **Free-from type**
  - Harder to parse
  - Easy to extend
- **Mainly represented in binary format**

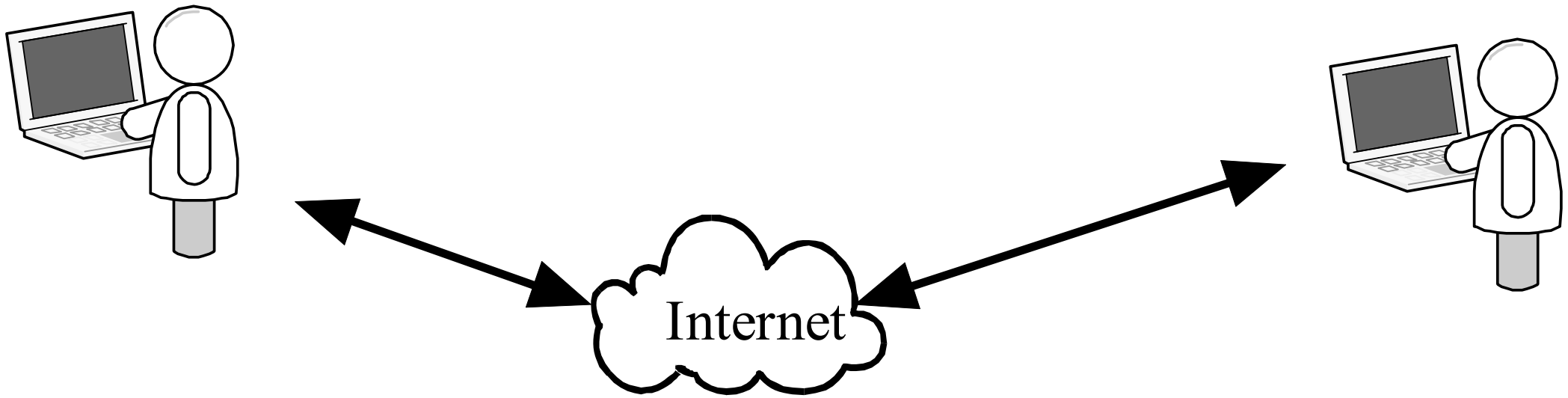
Protocol Headers

- **Fixed:**
  - Addresses (layer addresses and payload type)
  - Payload data
  - Control data
  - Header data
- **Options:**
  - Extended fixed data
  - Optional control data
  - Optional payload control
- **Payload:** Content not generally a header's concern
- **Trailer:** Optional field used for error control

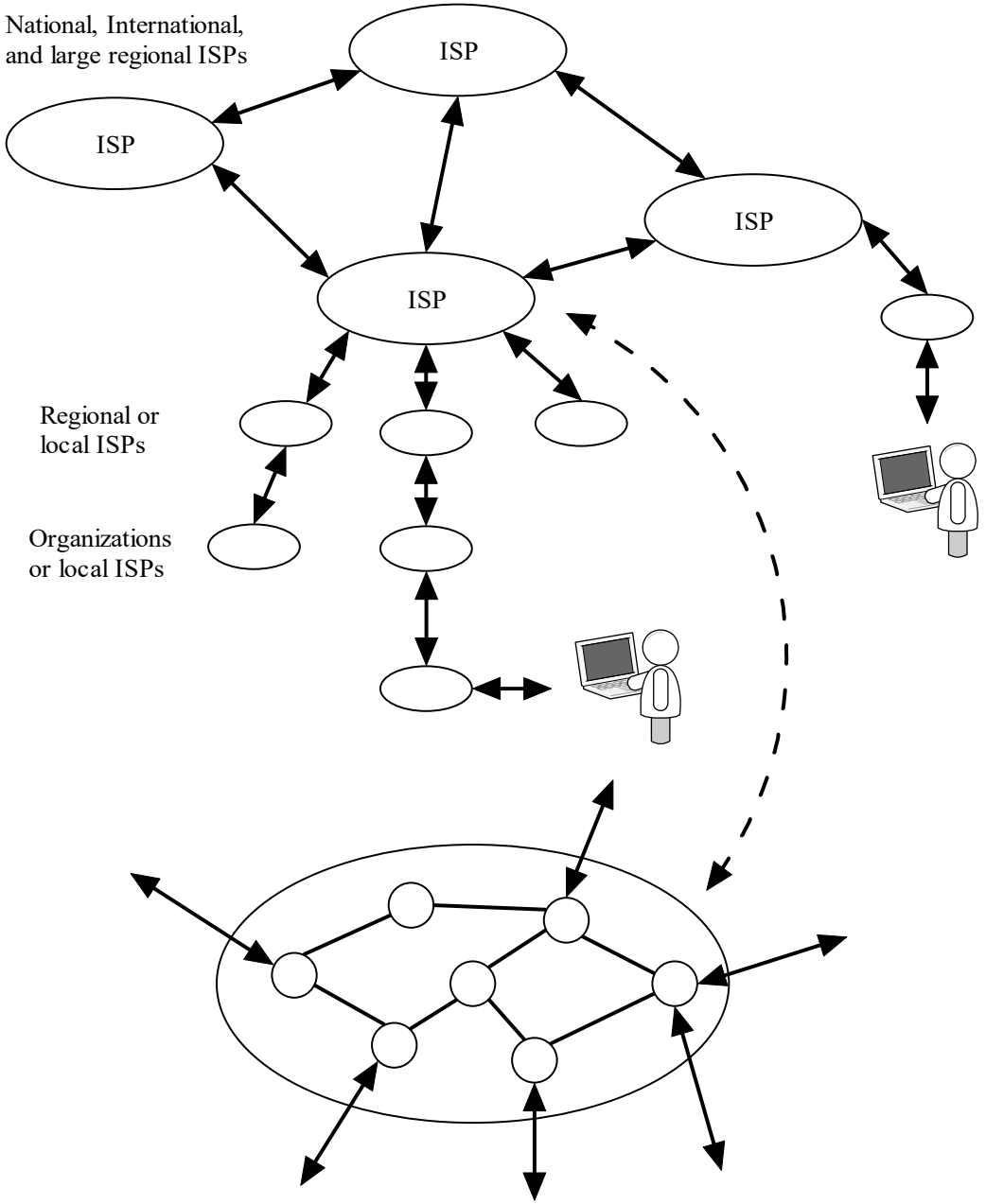
Fixed	Options	Payload	Trailer

Packet format

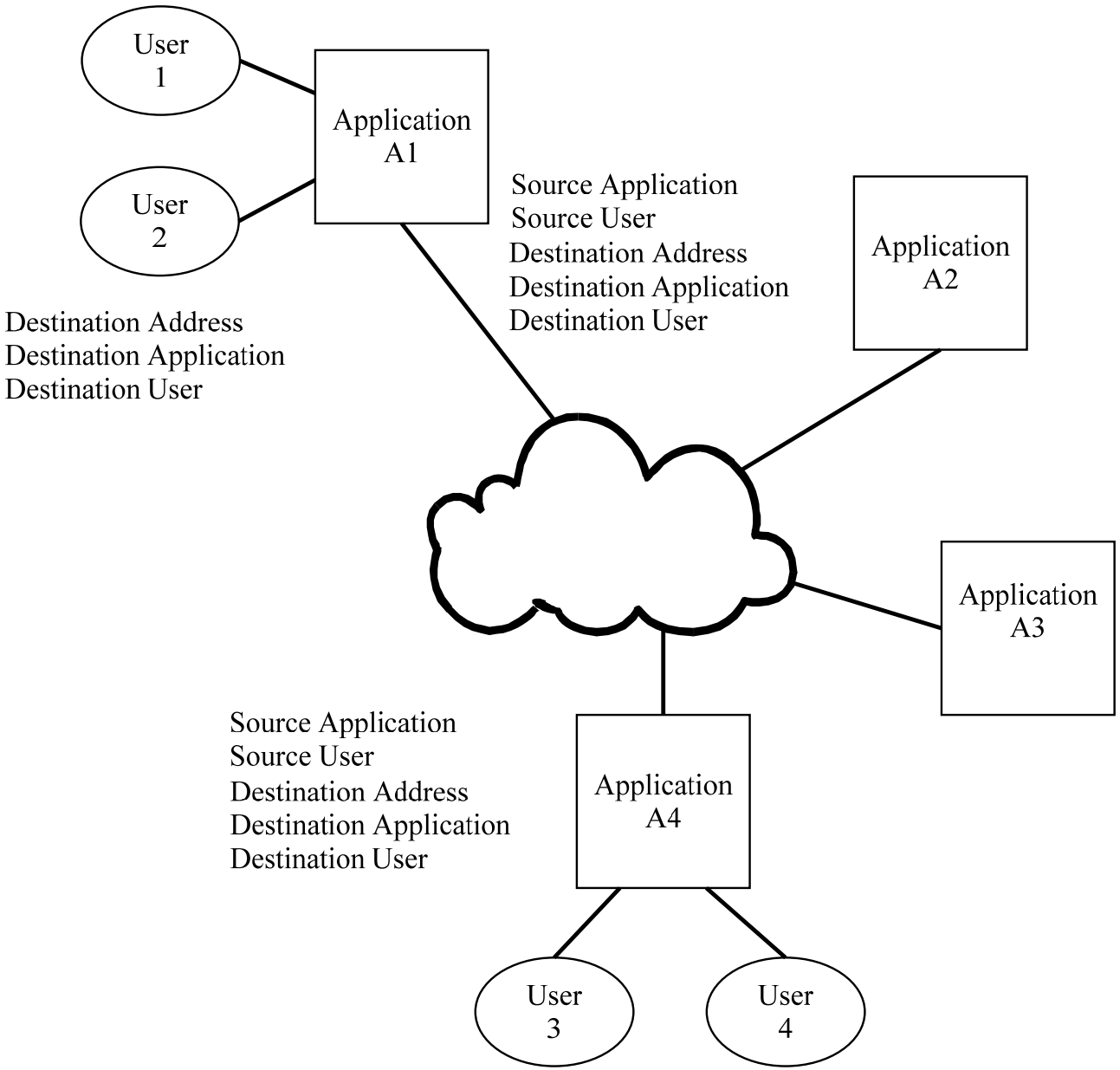
Users' View of the Internet



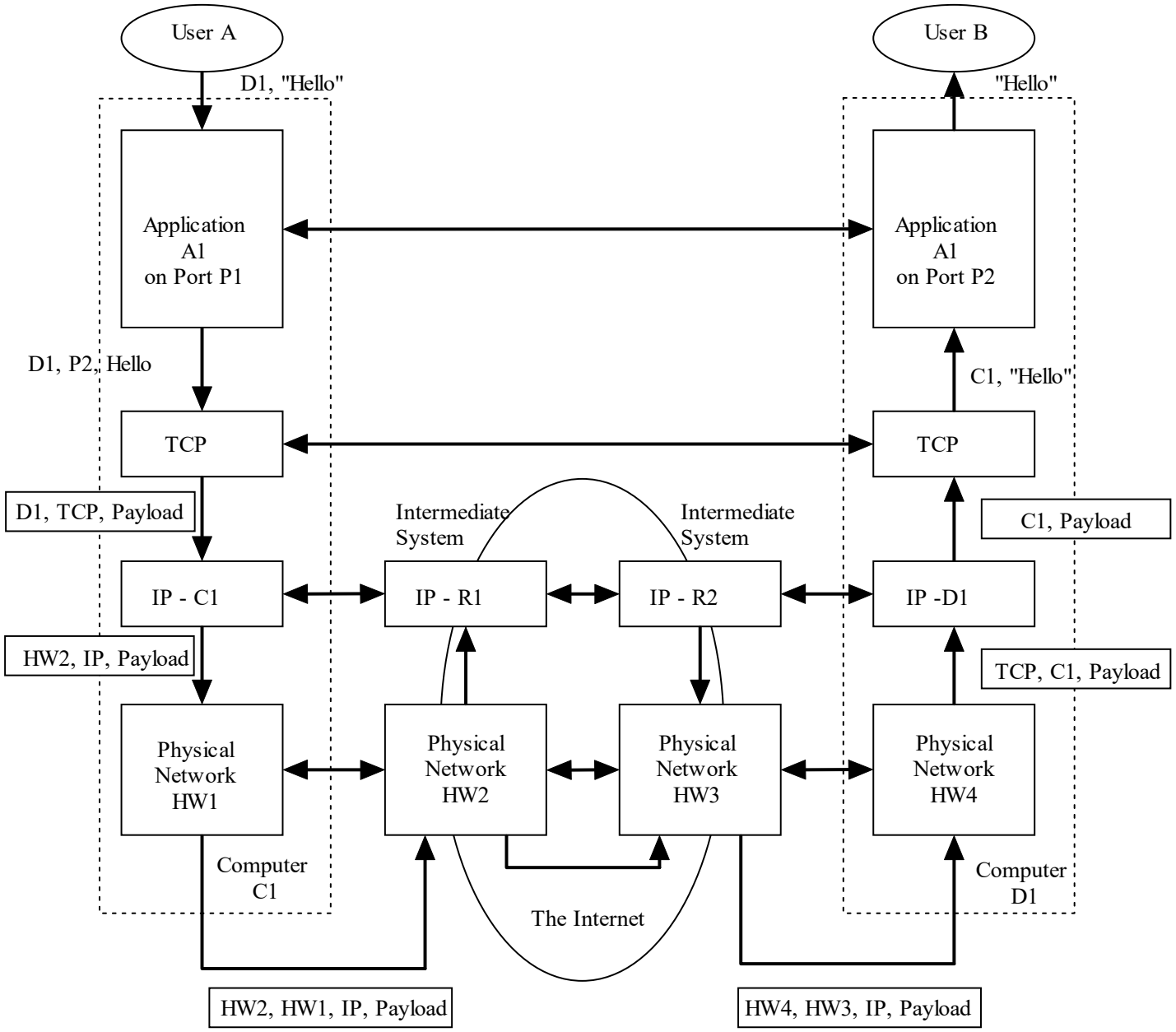
The Internet under the hood



Addressing on the Internet



Addressing on the Internet (more details)



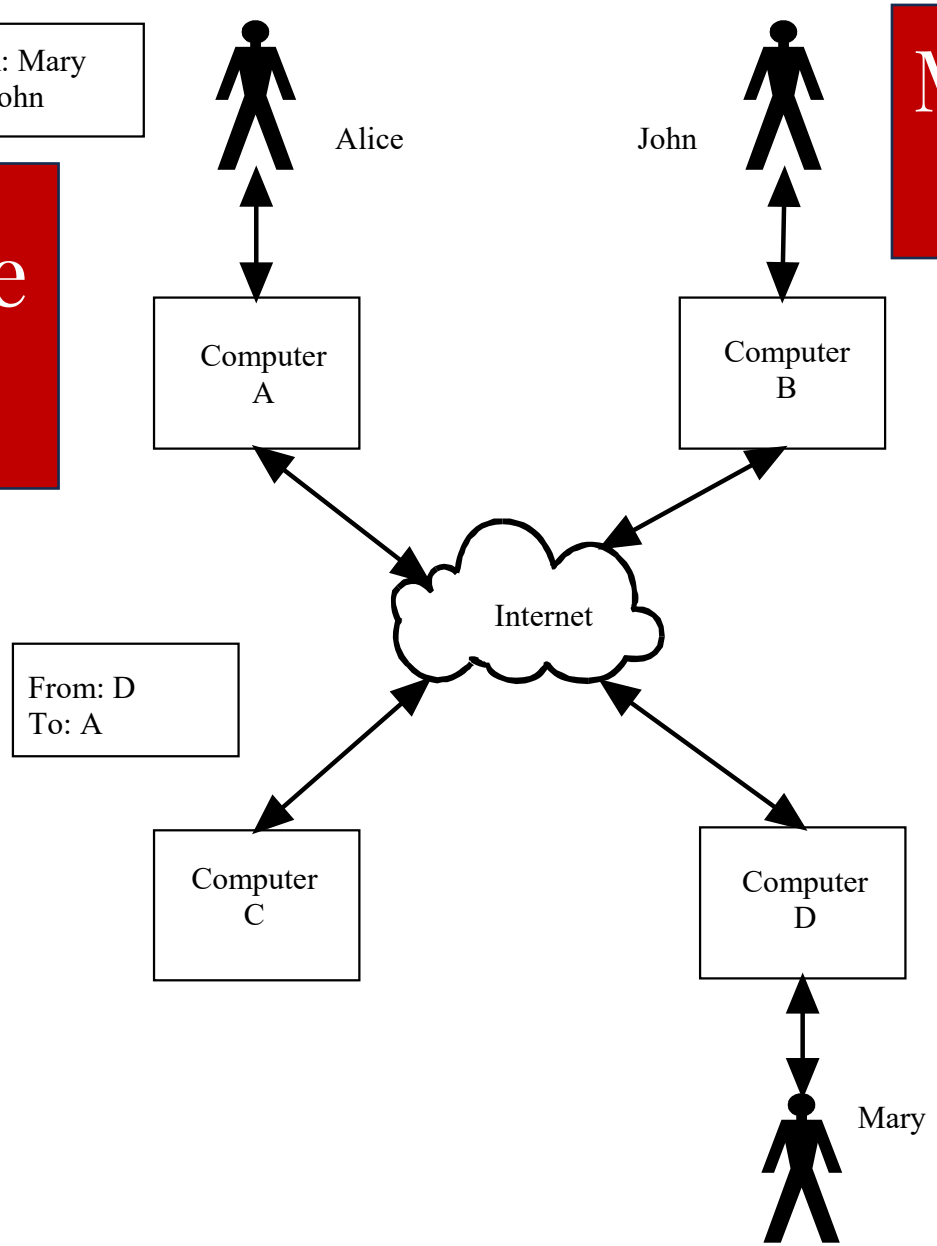
Address Spoofing

- **Who can generate the address?**

- Spoofing is the ability to change the address

- **Who can "see" (sniff) the traffic?**

Address Spoofing



The return message will get to Alice

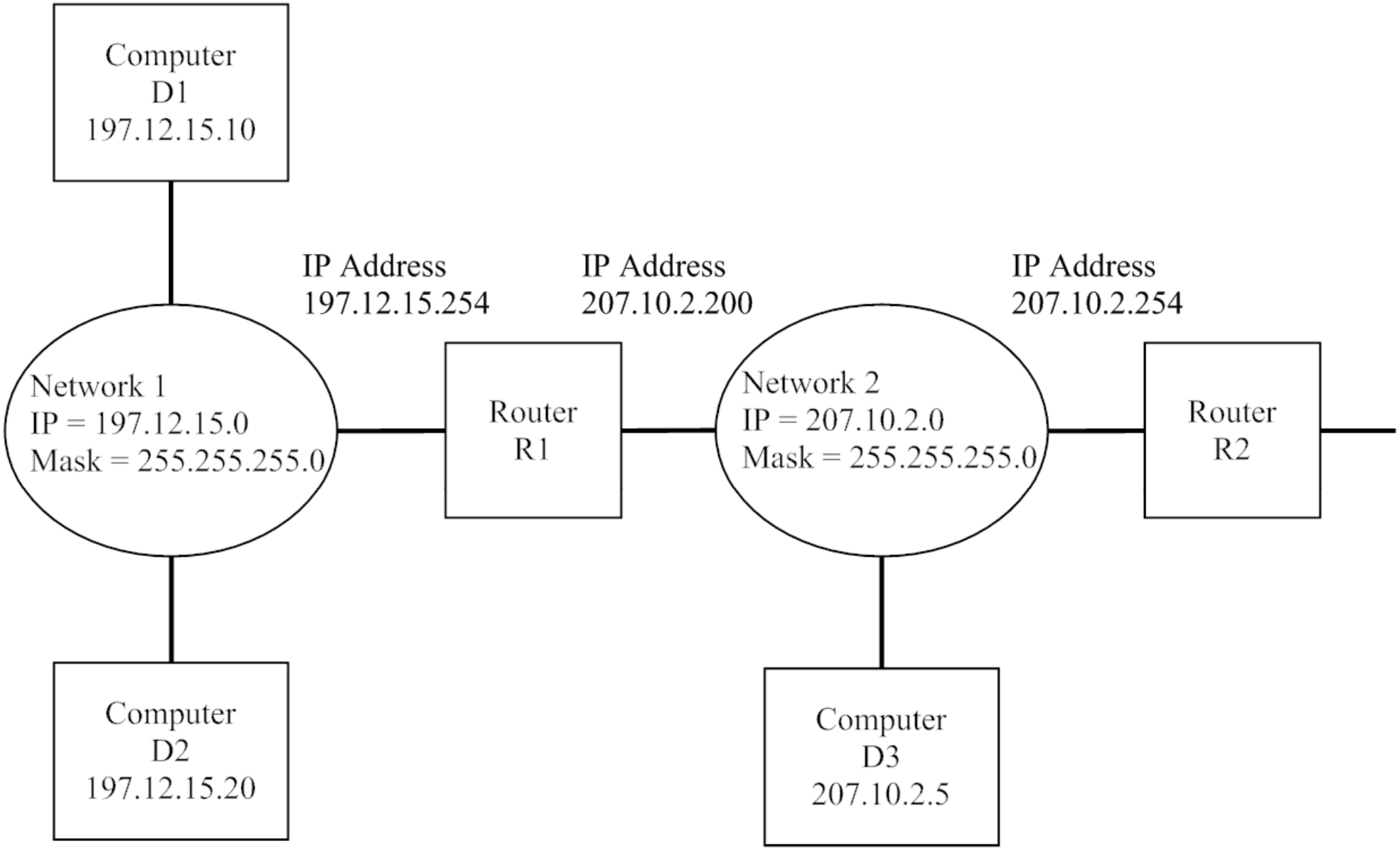
Message will get to John

IP Addresses

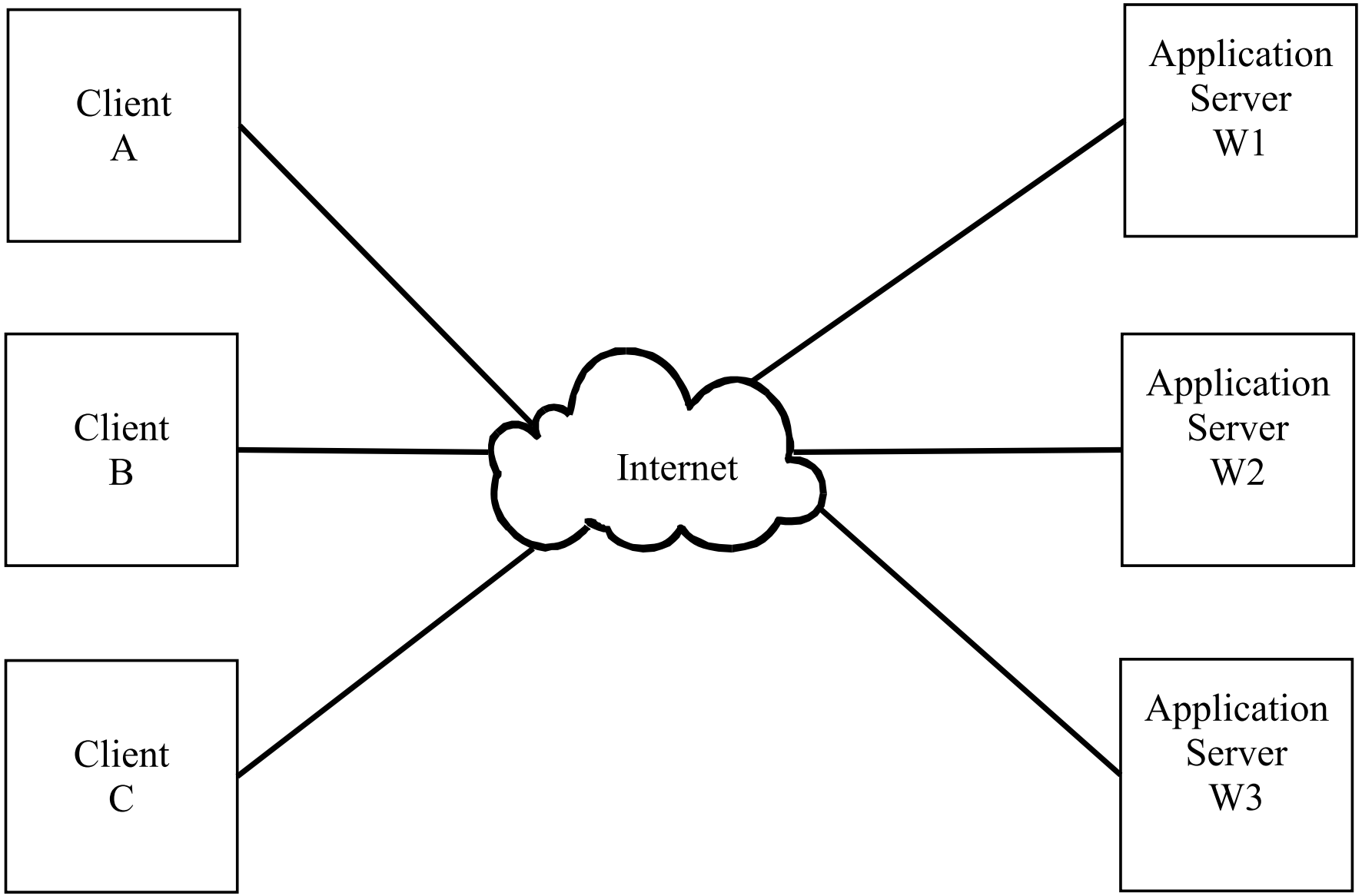
- **In Version 4, the IP address is 32 bits**
- Total IP address space is 4,294,967,296

- The IP address is written as a four-tuple where each tuple is in decimal and is separated by a "." (called a dot).
- When talking about an address, you pronounce the word dot. So 129.186.5.102 is pronounced 129 dot 186 dot 5 dot 102

Networks on the Internet



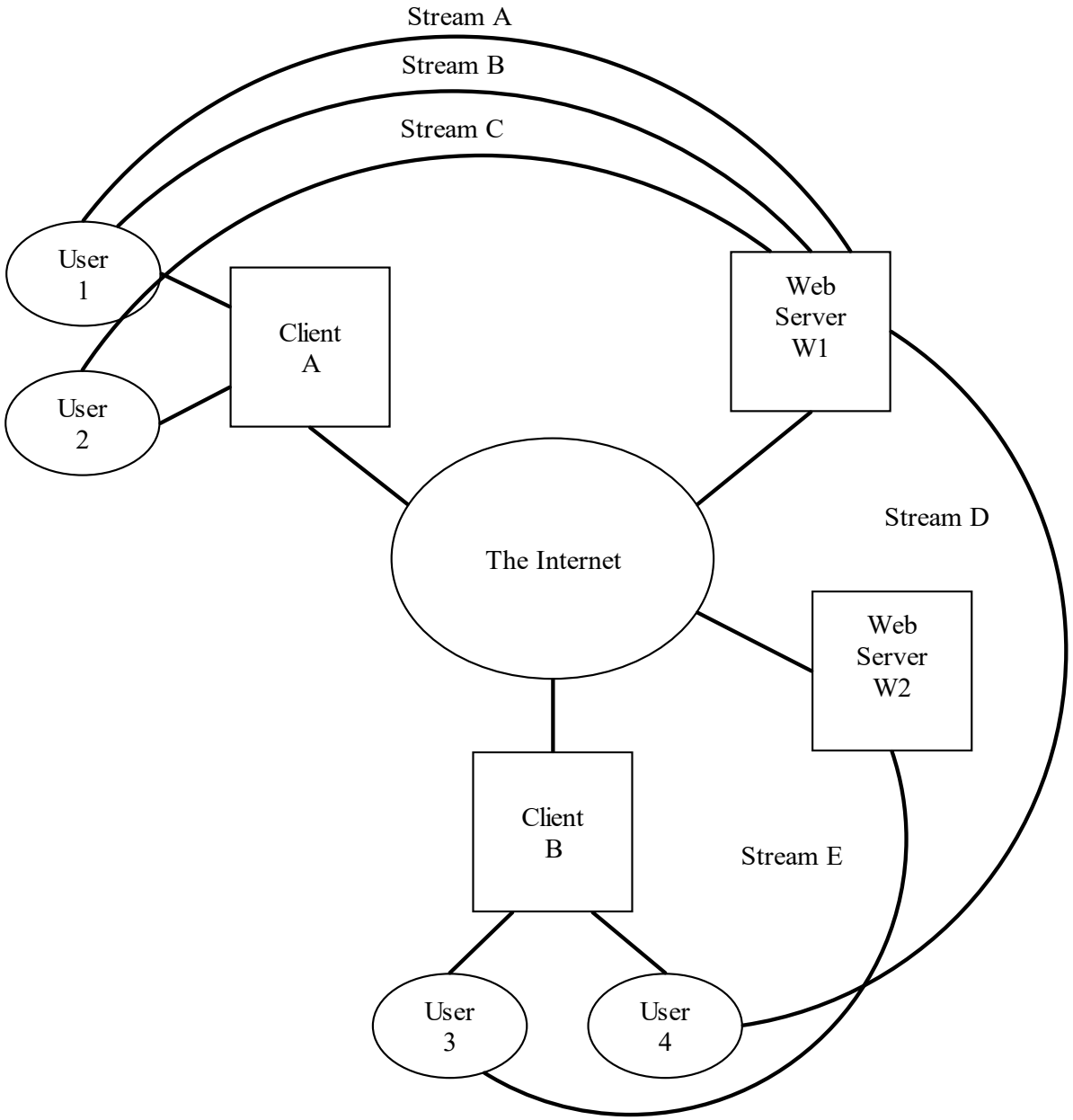
Client-Server Model



Client-Server Model

Packets from client to server	
Source IP	Client's IP address
Destination IP	Server's IP address
Source Port	Ephemeral port
Destination Port	Server's port number (often well known)
Packets from server to client	
Source IP	Server's IP address
Destination IP	Client's IP address
Source Port	Server's port number (often well known)
Destination Port	Ephemeral port

Real Client-Server Model



Real Client-Server Model

Stream	Source IP	Destination IP	Source Port	Destination Port
A	A	W1	Ephemeral A1	80
B	A	W1	Ephemeral A2	80
C	A	W1	Ephemeral A3	80
D	B	W1	Ephemeral B1	80
E	B	W2	Ephemeral B2	80

Routing

- All hosts and gateways store routing tables

- Each row in the route table contains:
  - Destination address or address range
  - Next hop for that destination address range
  - The physical interface to use for that address range. (i.e., which Ethernet card to use)

Example:

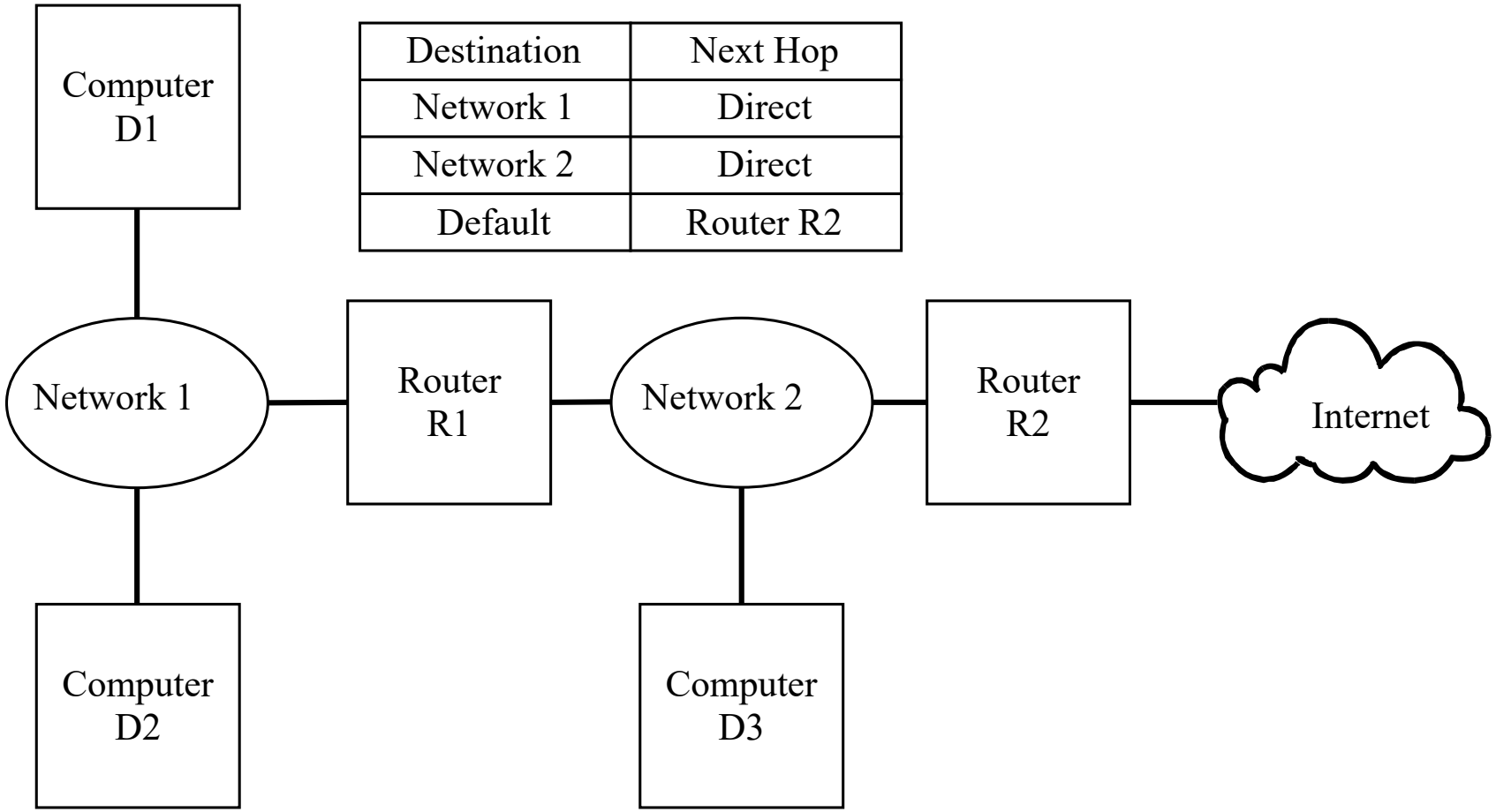
Destination	**Next**	**Interface**
129.186.4.0	129.186.5.254	en0

Dynamic vs Static Routing

- **Static**:
  - Tables are built at system configuration time
  - Used in small networks or networks with only one way out

- **Dynamic**:
  - Tables are modified based on network parameters
  - Used in larger networks with multiple paths

Routing

Destination	Next Hop
Network 1	Direct
Default	Router R1

Destination	Next Hop
Network 1	Direct
Network 2	Direct
Default	Router R2